Consistent Partial Least Squares: A cSEM tutorial

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Abstract

This tutorial article provides a comprehensive, step-by-step illustration of consistent partial least squares (PLSc) using the cSEM package in R. PLSc is a variant of partial least squares path modeling, a composite-based approach to structural equation modeling, that produces consistent parameter estimates for latent variable models. In particular, PLSc corrects construct correlations for attenuation using Dijkstra-Henseler's composite reliability coefficient ϱ_A. Building on the employee retention example introduced by Cheah, Sarstedt, Hair, and Ringle (2026), we demonstrate how to install and load cSEM, specify the model using lavaan-style syntax, estimate the PLSc-SEM model, evaluate measurement model quality (internal consistency reliability, convergent validity, and discriminant validity), and assess the structural model (collinearity, effect sizes, model fit, and path coefficients with bootstrap inference). We further contrast the PLSc-SEM results with standard PLS-SEM estimates and discuss practical considerations for using PLSc-SEM in empirical research.

Keywords: measurement error, bias correction, partial least squares structural equation modeling (PLS-SEM), factor score regression

1 Introduction

Structural equation modeling (SEM) is a versatile multivariate analysis framework used in the behavioral, social, medical, and management sciences to study complex relationships between constructs and their relations with observed variables (e.g., Kline, 2023; Marcoulides & Schumacker, 1996). To estimate structural equation models, various approaches have been proposed, including maximum-likelihood (Jöreskog, 1970), weighted least squares (Browne, 1974), factor score regression (Devlieger & Rosseel, 2017; Skrondal & Laake, 2001) and (integrated) generalized structured component analysis (Hwang et al., 2021; Hwang & Takane, 2004)

A composite-based estimator to SEM that gained popularity over the last decades in various fields of social sciences (Cook & Forzani, 2023), including marketing (Sarstedt et al., 2022), information systems research (Benitez, Henseler, Castillo, & Schuberth, 2020; Evermann & Rönkkö, 2023), human resource management (Ringle, Sarstedt, Mitchell, & Gudergan, 2020), and tourism research (Müller, Schuberth, & Henseler, 2018), is partial least squares path modeling (PLS-PM, Wold, 1982),also known as partial least squares structural equation modeling (PLS-SEM, Hair, Ringle, & Sarstedt, 2011). PLS-PM is a two-step estimation procedure (Schuberth, Zaza, & Henseler, 2023). In the first step, construct scores are obtained by the partial least squares algorithm. Subsequently, these construct scores are used to estimate the model parameters. Although PLS-PM is computationally efficient, it is known that in its original form to produce inconsistent estimates for models involving latent variables due to attenuation bias (e.g., Dijkstra, 1985; Hui & Wold, 1982; Rönkkö & Evermann, 2013; Schuberth, Rosseel, et al., 2023).

To address this limitation, consistent partial least squares (PLSc, Dijkstra, 2013; Dijkstra & Henseler, 2015a, 2015b; Dijkstra & Schermelleh-Engel, 2014) was introduced. PLSc is based on PLS-PM, but applies a correction for attenuation to obtain consistent estimates of reflective measurement models and path coefficient between latent variables. Additionally, PLSc has been extended to deal with ordinal variables (Schuberth, Henseler, & Dijkstra, 2018), randomly occurring outliers (Schamberger, Schuberth, Henseler, & Dijkstra, 2020) and multicollinearity issues (Jung & Park, 2018).

To apply PLS-PM and PLSc, various implementations have been developed (Venturini, Mehmetoglu, & Latan, 2023). They can be distinguished into: (i) proprietary software and (ii) open-source software. Proprietary software for conducting PLS-PM and PLSc includes SmartPLS (Ringle, Wende, & Becker, 2024), WarpPLS (Kock, 2025) and ADANCO (Henseler, 2025). On the other hand, open-source software alternatives include implementations in the statistical programming environment R (R Core Team, 2025) such as matrixpls (Rönkkö, 2022), SEMinR (Ray, Danks, & Calero Valdez, 2026), cSEM (Rademaker & Schuberth, 2020) and plspm (Sanchez, Trinchera, Russolillo, & Bertrand, 2025). Similarly, it includes implementations in python (Python Software Foundation, 2024) such as plspm (Humble, 2024). While proprietary software such as SmartPLS often shows high ease of use (Cheah, Magno, & Cassia, 2024), they conflict with the open science principles (UNESCO, 2021). For example, their codebase is not openly accessible of transparency (Morin et al., 2012), reproducibility (Ince, Hatton, & Graham-Cumming, 2012), and accessibility/inclusiveness (). Open-source software overcomes these limitations (Barba, 2022). Therefore, this tutorial presents the R package cSEM as a full-fledged open-source alternative to SmartPLS to apply PLS-PM and PLSc.

The remainder of the tutorial is structured as follows. Section 2 shows how to get started with R, introduces the employee retention model and data used for the illustration, and demonstrates how the model is specified. Section 3 covers the estimation of the model with the `csem()` function as well as the assessment of the measurement and the structural model. Section 4 presents the options to customize the PLS algorithm. Section 5 compares the PLSc results with those of alternative estimators. Section 6 highlights further functionality of the cSEM package. Finally, Section 7 concludes with a discussion.

2 cSEM R Illustration

2.1 Getting started with R

To use the cSEM package, users first need to install R and an integrated development environment (IDE) such as RStudio. For the installation, we refer to the official guides for R (`https://cran.r-project.org`) and RStudio (`https://posit.co/download/rstudio-desktop`).

When RStudio is opened for the first time, it displays four panes (Figure 1). The two most important ones are the *script editor* (upper left), where code is written and saved, and the *Console* (lower left), where the code is executed and results are printed. The *Environment* pane (upper right) lists the objects currently loaded into memory, and the fourth pane (lower right) provides access to files, plots, and help pages.

Although code can be typed directly into the console, it is good practice to work in a script, since a script can be saved, shared, and re-run, which is essential for reproducible analyses. To create a new script, click File → New File → R Script. The typical workflow is to write code in the editor and run the selected line(s) with `Ctrl + Enter`. Any line beginning with `#` is treated as a comment: it documents what the code does and is ignored when the code runs.

```
# This is a comment and is not executed.
1 + 1    # Code is executed; the result is printed in the Console.

[1] 2
```

Before cSEM can be used, it must be installed. The cSEM package is available on CRAN, so the most recent stable version can be installed with:

```
install.packages("cSEM")
```

Installation only needs to be done once per machine (rerun this line to update the package later). Moreover, development versions can be installed from GitHub: `https://github.com/FloSchuberth/cSEM`. After installation, the package must be loaded at the start of every new R session before its functions become available:

```
library(cSEM)


Attache Paket:  'cSEM'
Das folgende Objekt ist maskiert 'package:stats':
      predict
Das folgende Objekt ist maskiert 'package:grDevices':
      savePlot
```

With cSEM installed and loaded, we can now import the data and specify the model, as described in the following sections.

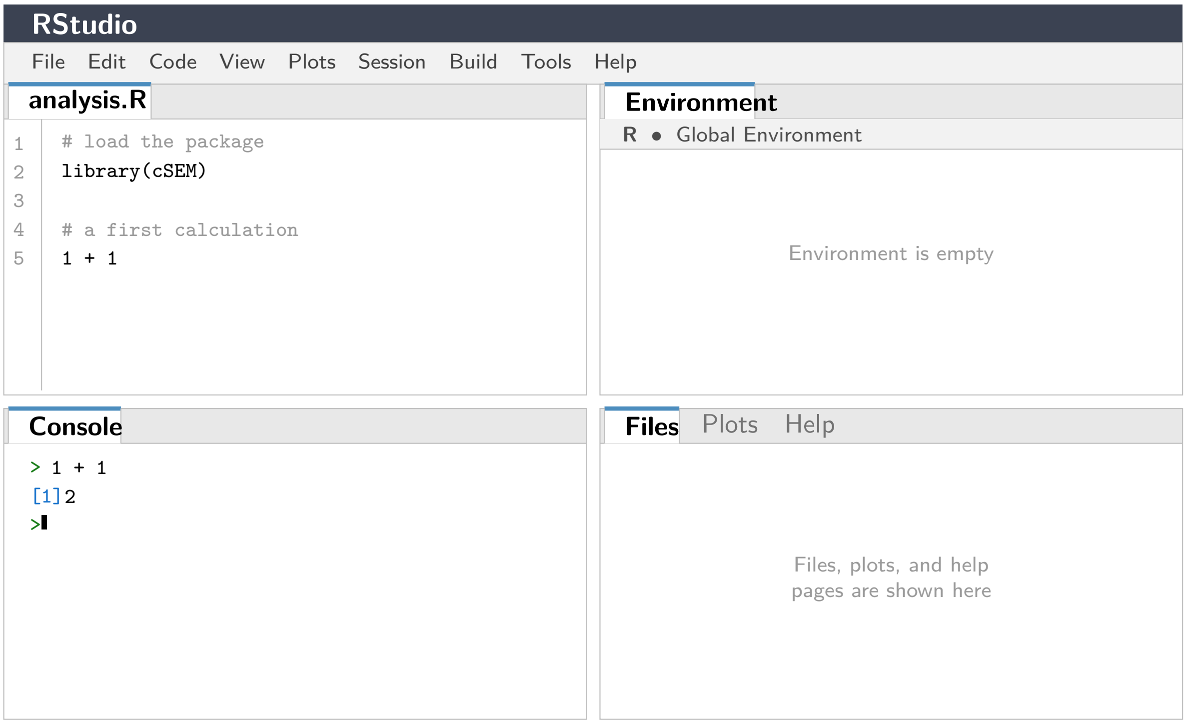

Figure 1: The four panes of the RStudio interface: the script editor (upper left), the *Console* (lower left), the *Environment* pane (upper right), and the files/plots/help pane (lower right).

2.2 Model and Data

The cSEM package (Rademaker & Schuberth, 2020) offers a variety of composite-based approaches to SEM including PLS-PM and PLSc. Following Cheah et al. (2026), our illustration of the application of PLSc using cSEM draws on the employee retention model introduced by Hair, Black, Babin, and Anderson (2019, Chapter 13). The model explains the effects of organizational commitment (OC) and job satisfaction (JS) on employees' staying intentions (SI), with work environment perceptions (EP) and attitudes toward coworkers (AC) as exogenous antecedent constructs. Five constructs are reflectively measured by a total of 21 indicators. A detailed description of the indicators can be found in Cheah et al. (2026). The model is presented in Figure 2.

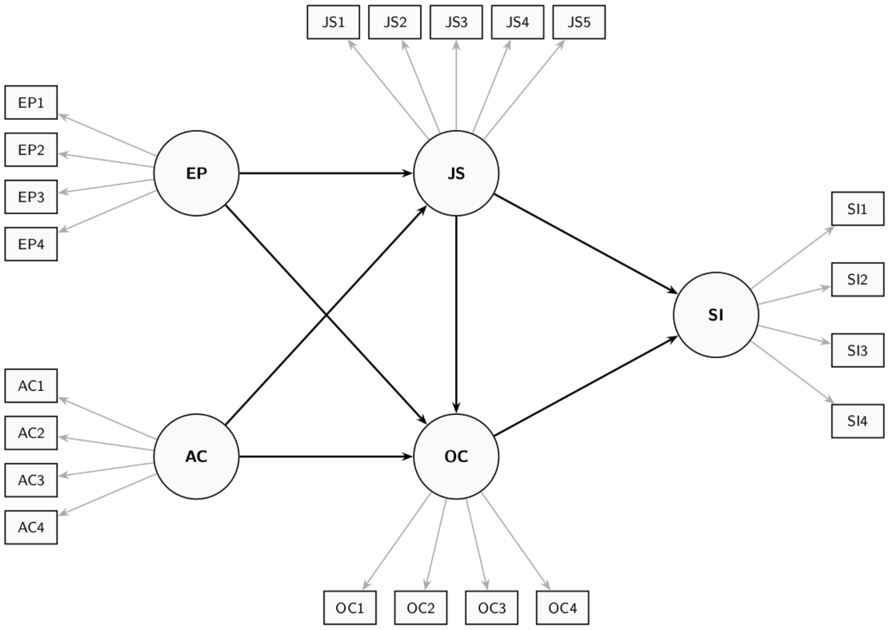

Figure 2: Employee Retention Model

The dataset used for model estimation is publicly available from the SmartPLS website: `https://www.smartpls.com/documentation/sample-projects/employee-retention`. The synthetic dataset comprises $N = 400$ observations from HBAT Industries and was generated to satisfy the assumptions of factor-based SEM, including multivariate normality. For this tutorial, we assume the data has been downloaded as a CSV file. After downloading the dataset, it needs to be loaded into the R Environment. The Employee Retention dataset is stored as an CSV File using ';' as the separator, thus it can be loaded into R using following command:

```
# Import dataset
HBAT_SEM <- read.csv("data/HBAT_SEM.csv", sep=";")

# Select relevant variables
HBAT_SEM_rel <- HBAT_SEM[, c("AC1", "AC2", "AC3", "AC4",
                   "EP1", "EP2", "EP3", "EP4",
                   "JS1", "JS2", "JS3", "JS4", "JS5",
                   "OC1", "OC2", "OC3", "OC4",
                   "SI1", "SI2", "SI3", "SI4")]

# show dimensions of the relevant dataset
dim(HBAT_SEM_rel)

[1] 400  21

# count missing values per variable
colSums(is.na(HBAT_SEM_rel))

AC1 AC2 AC3 AC4 EP1 EP2 EP3 EP4 JS1 JS2 JS3 JS4 JS5 OC1 OC2 OC3 OC4 SI1 SI2 SI3
  0   0   0   1   0   0   0   1   0   0   0   0   0   0   0   0   0   0   0   0
SI4
  0
```

Since cSEM can only handle complete datasets, i.e., datasets without any missing values, we check the number of missing values per variable using the `is.na()` function. The output shows that the variables AC4 and EP4 each contain two missing values. Before we can continue, we thus need to either apply imputation methods or delete the incomplete observations. Deleting incomplete observations can be done via following command:

```
# filter the dataset, only select complete cases
HBAT_SEM_rel <- HBAT_SEM_rel[complete.cases(HBAT_SEM_rel),]

#  the number of rows changed from 400 to 398
dim(HBAT_SEM_rel)

[1] 398  21
```

Since two observations contain missing values, all analyses in the remainder of this tutorial are based on the $N = 398$ complete observations. When using their own data, users must ensure it is stored as a data frame or a matrix before passing it to cSEM functions; if necessary, the data can be converted with `as.data.frame()` or `as.matrix()`.

2.3 Model specification

In cSEM, models are specified using the lavaan model syntax. The lavaan package does not need to be installed to use the syntax. In particular, the `=~` and the `<~` operators are used to define the relations between the constructs and their indicators. Specifically, the `=~` operator is used to specify reflective measurement models, while the `<~` is used to specify composite models (see , for an explanation of reflective and composite models). In addition, the `~` operator is used to specify the relations among constructs. In case, an observed variable should be directly accounted for in the structural model, it must be specified as a single indicator construct, as common in PLS-PM (). The cSEM specification of the retention model illustrated in Figure 2 is as follows:

```
model <- "
  # Measurement model
  AC =~ AC1 + AC2 + AC3 + AC4
  EP =~ EP1 + EP2 + EP3 + EP4
  JS =~ JS1 + JS2 + JS3 + JS4 + JS5
  OC =~ OC1 + OC2 + OC3 + OC4
  SI =~ SI1 + SI2 + SI3 + SI4

  # Structural model
  JS ~ EP + AC
  OC ~ EP + AC + JS
  SI ~ JS + OC
"
```

3 Estimating the model with cSEM

3.1 Running the estimation

The central function is `csem()`. The minimal input required for the `csem()` function is a dataset and a model:

.data A data frame or matrix with column names matching the indicator names used in the model description. Possible column types of data proved are: "`logical`", "`numeric`" ("`double` or "`integer`"), "`factor`" ("`ordered` and/or "`unordered`"), "`character`" or a mix of several types.

.model A model string written in `lavaan` syntax, with `=~` for measurement relations and `~` for structural (regression) relations.

By default weights are estimated using the partial leasts sqaures path modeling algorithm ("PLS-PM").

To ultimately estimate with the csem function, we call it and save the results as `res_csem`:

```
# PLSc-SEM estimation
res_csem <- csem(
  .data  = HBAT_SEM_rel,
  .model = model
)
```

To obtain a comprehensive overview of all results, including loadings, path coefficients, reliability measures, and R^2 values, use:

```
# Full results summary
summary_res <- summarize(res_csem)
print(summary_res)


--------------------------------------------------------------------------------
----------------------------------- Overview -----------------------------------

 General information:
 ------------------------
 Estimation status                  = Ok
 Number of observations             = 398
 Weight estimator                   = PLS-PM
 Inner weighting scheme             = "path"
 Type of indicator correlation      = Pearson
 Path model estimator               = OLS
 Second-order approach              = NA
 Type of path model                 = Linear
 Disattenuated                      = Yes (PLSc)

 Construct details:
 ------------------
 Name  Modeled as     Order         Mode

 EP    Common factor  First order   "modeA"
 AC    Common factor  First order   "modeA"
 JS    Common factor  First order   "modeA"
 OC    Common factor  First order   "modeA"
 SI    Common factor  First order   "modeA"

----------------------------------- Estimates ----------------------------------

Estimated path coefficients:
```

```
=============================
  Path       Estimate  Std. error   t-stat.   p-value
  JS ~ EP      0.2448          NA        NA        NA
  JS ~ AC     -0.0033          NA        NA        NA
  OC ~ EP      0.4288          NA        NA        NA
  OC ~ AC      0.1726          NA        NA        NA
  OC ~ JS      0.0961          NA        NA        NA
  SI ~ JS      0.1312          NA        NA        NA
  SI ~ OC      0.4900          NA        NA        NA

Estimated loadings:
===================
  Loading      Estimate  Std. error   t-stat.   p-value
  EP =~ EP1      0.6341          NA        NA        NA
  EP =~ EP2      0.8787          NA        NA        NA
  EP =~ EP3      0.7678          NA        NA        NA
  EP =~ EP4      0.8231          NA        NA        NA
  AC =~ AC1      0.7691          NA        NA        NA
  AC =~ AC2      0.6765          NA        NA        NA
  AC =~ AC3      0.8216          NA        NA        NA
  AC =~ AC4      0.9934          NA        NA        NA
  JS =~ JS1      0.6248          NA        NA        NA
  JS =~ JS2      0.6965          NA        NA        NA
  JS =~ JS3      0.7182          NA        NA        NA
  JS =~ JS4      0.6319          NA        NA        NA
  JS =~ JS5      0.9004          NA        NA        NA
  OC =~ OC1      0.4252          NA        NA        NA
  OC =~ OC2      0.9575          NA        NA        NA
  OC =~ OC3      0.6859          NA        NA        NA
  OC =~ OC4      0.8566          NA        NA        NA
  SI =~ SI1      0.8170          NA        NA        NA
  SI =~ SI2      0.8705          NA        NA        NA
  SI =~ SI3      0.6778          NA        NA        NA
  SI =~ SI4      0.8959          NA        NA        NA

Estimated weights:
==================
  Weight       Estimate  Std. error   t-stat.   p-value
  EP <~ EP1      0.2425          NA        NA        NA
  EP <~ EP2      0.3361          NA        NA        NA
  EP <~ EP3      0.2937          NA        NA        NA
  EP <~ EP4      0.3148          NA        NA        NA
  AC <~ AC1      0.2707          NA        NA        NA
  AC <~ AC2      0.2381          NA        NA        NA
  AC <~ AC3      0.2892          NA        NA        NA
  AC <~ AC4      0.3496          NA        NA        NA
  JS <~ JS1      0.2223          NA        NA        NA
  JS <~ JS2      0.2478          NA        NA        NA
  JS <~ JS3      0.2556          NA        NA        NA
  JS <~ JS4      0.2249          NA        NA        NA
  JS <~ JS5      0.3204          NA        NA        NA
  OC <~ OC1      0.1741          NA        NA        NA
  OC <~ OC2      0.3920          NA        NA        NA
```

```
  OC <~ OC3      0.2808          NA        NA        NA
  OC <~ OC4      0.3507          NA        NA        NA
  SI <~ SI1      0.2882          NA        NA        NA
  SI <~ SI2      0.3071          NA        NA        NA
  SI <~ SI3      0.2391          NA        NA        NA
  SI <~ SI4      0.3161          NA        NA        NA

Estimated construct correlations:
=================================
  Correlation    Estimate  Std. error   t-stat.   p-value
  EP ~~ AC         0.2549          NA        NA        NA

------------------------------------ Effects -----------------------------------

Estimated total effects:
========================
  Total effect    Estimate  Std. error   t-stat.   p-value
  JS ~ EP           0.2448          NA        NA        NA
  JS ~ AC          -0.0033          NA        NA        NA
  OC ~ EP           0.4523          NA        NA        NA
  OC ~ AC           0.1722          NA        NA        NA
  OC ~ JS           0.0961          NA        NA        NA
  SI ~ EP           0.2538          NA        NA        NA
  SI ~ AC           0.0840          NA        NA        NA
  SI ~ JS           0.1783          NA        NA        NA
  SI ~ OC           0.4900          NA        NA        NA

Estimated indirect effects:
===========================
  Indirect effect    Estimate  Std. error   t-stat.   p-value
  OC ~ EP              0.0235          NA        NA        NA
  OC ~ AC             -0.0003          NA        NA        NA
  SI ~ EP              0.2538          NA        NA        NA
  SI ~ AC              0.0840          NA        NA        NA
  SI ~ JS              0.0471          NA        NA        NA
________________________________________________________________________________
```

```
# For a more compact overview
summary_res$Estimates$Loading_estimates
```

```
        Name Construct_type  Estimate Std_err t_stat p_value
1  EP =~ EP1  Common factor 0.6341011      NA     NA      NA
2  EP =~ EP2  Common factor 0.8786819      NA     NA      NA
3  EP =~ EP3  Common factor 0.7678021      NA     NA      NA
4  EP =~ EP4  Common factor 0.8230691      NA     NA      NA
5  AC =~ AC1  Common factor 0.7691299      NA     NA      NA
6  AC =~ AC2  Common factor 0.6765259      NA     NA      NA
7  AC =~ AC3  Common factor 0.8215860      NA     NA      NA
8  AC =~ AC4  Common factor 0.9933858      NA     NA      NA
9  JS =~ JS1  Common factor 0.6248337      NA     NA      NA
10 JS =~ JS2  Common factor 0.6964587      NA     NA      NA
11 JS =~ JS3  Common factor 0.7182274      NA     NA      NA
12 JS =~ JS4  Common factor 0.6319045      NA     NA      NA
13 JS =~ JS5  Common factor 0.9004003      NA     NA      NA
```

```
14 OC =~ OC1  Common factor 0.4252105      NA     NA      NA
15 OC =~ OC2  Common factor 0.9574682      NA     NA      NA
16 OC =~ OC3  Common factor 0.6858765      NA     NA      NA
17 OC =~ OC4  Common factor 0.8565670      NA     NA      NA
18 SI =~ SI1  Common factor 0.8170099      NA     NA      NA
19 SI =~ SI2  Common factor 0.8705194      NA     NA      NA
20 SI =~ SI3  Common factor 0.6778349      NA     NA      NA
21 SI =~ SI4  Common factor 0.8959341      NA     NA      NA

summary_res$Estimates$Path_estimates

     Name Construct_type     Estimate Std_err t_stat p_value
1 JS ~ EP  Common factor  0.244824960      NA     NA      NA
2 JS ~ AC  Common factor -0.003282073      NA     NA      NA
3 OC ~ EP  Common factor  0.428781485      NA     NA      NA
4 OC ~ AC  Common factor  0.172554413      NA     NA      NA
5 OC ~ JS  Common factor  0.096069429      NA     NA      NA
6 SI ~ JS  Common factor  0.131226588      NA     NA      NA
7 SI ~ OC  Common factor  0.490013776      NA     NA      NA
```

Note that the standard errors, t-values, and p-values in the `summarize()` output are reported as `NA`. The reason is that a plain `csem()` call computes point estimates only. To conduct statistical inference, the standard errors of the estimates must be obtained by a resampling procedure such as the bootstrap, which is introduced in Section 3.2.

3.2 Assessing the measurement model

The next step involves assessing the measurement model results, according the guidelines described in !!!Benites!!!. The `assess()` function in cSEM provides all relevant quality criteria:

```
# Compute all quality criteria
quality <- assess(res_csem)

Warning:  The following warning occured in the calculateHTMT() function:
Intra-block and inter-block correlations between indicators must be either all-positive
or all-negative.
Warning:  The following warning occured in the calculateHTMT() function:
Intra-block and inter-block correlations between indicators must be either all-positive
or all-negative.
Warning in log(x[[3]]):  NaNs wurden erzeugt

# Internal consistency reliability
quality

--------------------------------------------------------------------------------

 Construct        AVE           R2          R2_adj
 EP             0.6103          NA            NA
 AC             0.6778          NA            NA
 JS             0.5203        0.0595        0.0548
 OC             0.5754        0.2826        0.2772
 SI             0.6719        0.2845        0.2808
```

```
-------------- Common (internal consistency) reliability estimates -------------

 Construct Cronbachs_alpha   Joereskogs_rho   Dijkstra-Henselers_rho_A
 EP        0.8596           0.8607                0.8717
 AC        0.8936           0.8919                0.9105
 JS        0.8450           0.8417                0.8568
 OC        0.8328           0.8344                0.8879
 SI        0.8890           0.8902                0.8989

----------- Alternative (internal consistency) reliability estimates -----------

 Construct       RhoC         RhoC_mm    RhoC_weighted
 EP             0.8607        0.8557        0.8717
 AC             0.8919        0.8765        0.9105
 JS             0.8417        0.8267        0.8568
 OC             0.8344        0.8029        0.8879
 SI             0.8902        0.8860        0.8989

 Construct  RhoC_weighted_mm     RhoT      RhoT_weighted
 EP             0.8717        0.8596        0.8522
 AC             0.9105        0.8936        0.8923
 JS             0.8568        0.8450        0.8427
 OC             0.8879        0.8328        0.7954
 SI             0.8989        0.8890        0.8830

---------------------------- Distance and fit measures --------------------------

 Geodesic distance             = 0.3252628
 Squared Euclidean distance    = 1.150488
 ML distance                   = 1.89721

 Chi_square       = 753.1923
 Chi_square_df    = 4.161284
 CFI              = 0.8647056
 CN               = 113.476
 GFI              = 0.7555901
 IFI              = 0.865627
 NFI              = 0.830333
 NNFI             = 0.8430286
 RMSEA            = 0.08923526
 RMS_theta        = 0.04595267
 SRMR             = 0.07057243

 Degrees of freedom       = 181

--------------------------- Model selection criteria ---------------------------

 Construct        AIC          AICc          AICu
 JS           -19.4331      380.6687      -16.4217
 OC           -125.2111     274.9420      -121.1909
 SI           -128.2165     271.8852      -125.2052

 Construct        BIC           FPE           GM
```

```
 JS             -7.4737       0.9523       415.1565
 OC           -109.2653      0.7301       420.5491
 SI           -116.2572      0.7246       503.0830

 Construct         HQ            HQc       Mallows_Cp
 JS           -14.6961     -14.5595       5.1971
 OC           -118.8951    -118.6760      6.6033
 SI           -123.4796    -123.3430      93.1236

----------------------- Variance inflation factors (VIFs) ----------------------

  Dependent construct: 'JS'

 Independent construct    VIF value
 EP                        1.0695
 AC                        1.0695

  Dependent construct: 'OC'

 Independent construct    VIF value
 EP                        1.1333
 AC                        1.0695
 JS                        1.0633

  Dependent construct: 'SI'

 Independent construct    VIF value
 JS                        1.0465
 OC                        1.0465

-------------------------- Effect sizes (Cohen's f^2) --------------------------

  Dependent construct: 'JS'

 Independent construct       f^2
 EP                        0.0596
 AC                        0.0000

  Dependent construct: 'OC'

 Independent construct       f^2
 EP                        0.2262
 AC                        0.0388
 JS                        0.0121

  Dependent construct: 'SI'

 Independent construct       f^2
 JS                        0.0230
 OC                        0.3206

----------------------- Discriminant validity assessment -----------------------
```

```
 Heterotrait-monotrait ratio of correlations matrix (HTMT matrix)

          EP         AC        JS        OC SI
EP 1.0000000 0.00000000 0.0000000 0.0000000  0
AC 0.2555939 1.00000000 0.0000000 0.0000000  0
JS 0.2437979 0.05239146 1.0000000 0.0000000  0
OC 0.4939933 0.27449892 0.2086788 1.0000000  0
SI 0.5672796 0.30936763 0.2313363 0.5005239  1


 Advanced heterotrait-monotrait ratio of correlations matrix (HTMT2 matrix)

          EP        AC        JS        OC SI
EP 1.0000000 0.0000000 0.0000000 0.0000000  0
AC 0.2529206 1.0000000 0.0000000 0.0000000  0
JS 0.2376530       NaN 1.0000000 0.0000000  0
OC 0.4712082 0.2514075 0.1977162 1.0000000  0
SI 0.5661731 0.3083707 0.2219788 0.4705377  1


 Fornell-Larcker matrix

           EP          AC          JS         OC         SI
EP 0.61028217 0.064998129 0.059530244 0.24622791 0.31507449
AC 0.06499813 0.677766756 0.003496994 0.08268627 0.09452102
JS 0.05953024 0.003496994 0.520269313 0.04447507 0.05502128
OC 0.24622791 0.082686268 0.044475067 0.57542078 0.26800117
SI 0.31507449 0.094521019 0.055021276 0.26800117 0.67186684


------------------------------------ Effects -----------------------------------

Estimated total effects:
========================
  Total effect    Estimate  Std. error   t-stat.   p-value
  JS ~ EP           0.2448          NA        NA        NA
  JS ~ AC          -0.0033          NA        NA        NA
  OC ~ EP           0.4523          NA        NA        NA
  OC ~ AC           0.1722          NA        NA        NA
  OC ~ JS           0.0961          NA        NA        NA
  SI ~ EP           0.2538          NA        NA        NA
  SI ~ AC           0.0840          NA        NA        NA
  SI ~ JS           0.1783          NA        NA        NA
  SI ~ OC           0.4900          NA        NA        NA

Estimated indirect effects:
===========================
  Indirect effect    Estimate  Std. error   t-stat.   p-value
  OC ~ EP              0.0235          NA        NA        NA
  OC ~ AC             -0.0003          NA        NA        NA
  SI ~ EP              0.2538          NA        NA        NA
  SI ~ AC              0.0840          NA        NA        NA
  SI ~ JS              0.0471          NA        NA        NA
```

```
--------------------------------------------------------------------------------

# Indicator loadings
summary_res$Estimates$Loading_estimates

        Name Construct_type  Estimate Std_err t_stat p_value
1  EP =~ EP1  Common factor 0.6341011      NA     NA      NA
2  EP =~ EP2  Common factor 0.8786819      NA     NA      NA
3  EP =~ EP3  Common factor 0.7678021      NA     NA      NA
4  EP =~ EP4  Common factor 0.8230691      NA     NA      NA
5  AC =~ AC1  Common factor 0.7691299      NA     NA      NA
6  AC =~ AC2  Common factor 0.6765259      NA     NA      NA
7  AC =~ AC3  Common factor 0.8215860      NA     NA      NA
8  AC =~ AC4  Common factor 0.9933858      NA     NA      NA
9  JS =~ JS1  Common factor 0.6248337      NA     NA      NA
10 JS =~ JS2  Common factor 0.6964587      NA     NA      NA
11 JS =~ JS3  Common factor 0.7182274      NA     NA      NA
12 JS =~ JS4  Common factor 0.6319045      NA     NA      NA
13 JS =~ JS5  Common factor 0.9004003      NA     NA      NA
14 OC =~ OC1  Common factor 0.4252105      NA     NA      NA
15 OC =~ OC2  Common factor 0.9574682      NA     NA      NA
16 OC =~ OC3  Common factor 0.6858765      NA     NA      NA
17 OC =~ OC4  Common factor 0.8565670      NA     NA      NA
18 SI =~ SI1  Common factor 0.8170099      NA     NA      NA
19 SI =~ SI2  Common factor 0.8705194      NA     NA      NA
20 SI =~ SI3  Common factor 0.6778349      NA     NA      NA
21 SI =~ SI4  Common factor 0.8959341      NA     NA      NA
```

The results in Table **??** indicate that all constructs meet the commonly accepted thresholds. All Cronbach's α and ρ_{A} values exceed 0.70, and all AVE values exceed 0.50, supporting internal consistency reliability and convergent validity. While some outer loadings (e.g., OC1 = 0.427) fall below the recommended threshold of 0.708, these indicators are retained because the corresponding constructs exhibit adequate reliability and AVE, and their retention preserves content validity (Hair, Hult, Ringle, & Sarstedt, 2026).

Discriminant validity is assessed using the heterotrait-monotrait ratio of correlations (HTMT; Henseler, Ringle, & Sarstedt, 2015). In cSEM, HTMT values are available via `assess()`:

```
# HTMT values
quality$HTMT

$htmts
          EP         AC        JS        OC SI
EP 1.0000000 0.00000000 0.0000000 0.0000000  0
AC 0.2555939 1.00000000 0.0000000 0.0000000  0
JS 0.2437979 0.05239146 1.0000000 0.0000000  0
OC 0.4939933 0.27449892 0.2086788 1.0000000  0
SI 0.5672796 0.30936763 0.2313363 0.5005239  1

$quantiles
NULL

$nr_admissibles
NULL
```

```r
# Bootstrap confidence intervals for HTMT
res_csem_boot <- csem(
  .data              = HBAT_SEM_rel,
  .model             = model,
  .approach_weights = "PLS-PM",
  .disattenuate      = TRUE,
  .resample_method  = "bootstrap",
  .R                 = 10000,
  .seed              = 42
)
```

```
Warning:  Paket 'future' wurde unter R Version 4.5.3 erstellt
```

```r
# Extract HTMT inference
htmt_infer <- infer(res_csem_boot)

htmt_infer
```

```
$Path_estimates
$Path_estimates$mean
    JS ~ EP      JS ~ AC      OC ~ EP      OC ~ AC      OC ~ JS      SI ~ JS 
 0.25498139 -0.00699852  0.43417417  0.18400708  0.10066826  0.13183657 
    SI ~ OC 
 0.49777868 

$Path_estimates$sd
   JS ~ EP    JS ~ AC    OC ~ EP    OC ~ AC    OC ~ JS    SI ~ JS    SI ~ OC 
0.05238688 0.06081747 0.05735791 0.04732706 0.05289946 0.04685520 0.05095048 

$Path_estimates$bias
      JS ~ EP       JS ~ AC       OC ~ EP       OC ~ AC       OC ~ JS 
 0.0101564300 -0.0037164466  0.0053926890  0.0114526664  0.0045988304 
      SI ~ JS       SI ~ OC 
 0.0006099818  0.0077649037 

$Path_estimates$CI_standard_z
       JS ~ EP    JS ~ AC   OC ~ EP    OC ~ AC     OC ~ JS    SI ~ JS   SI ~ OC
95%L 0.1319921 -0.1187657 0.3109694 0.06834241 -0.01221044 0.03878211 0.3823878
95%U 0.3373449  0.1196344 0.5358082 0.25386108  0.19515164 0.22245111 0.5821100

$Path_estimates$CI_standard_t
       JS ~ EP    JS ~ AC   OC ~ EP    OC ~ AC     OC ~ JS    SI ~ JS   SI ~ OC
95%L 0.1316782 -0.1191302 0.3106256 0.06805876 -0.01252749 0.03850128 0.3820824
95%U 0.3376589  0.1199989 0.5361520 0.25414473  0.19546869 0.22273193 0.5824154

$Path_estimates$CI_percentile
       JS ~ EP    JS ~ AC   OC ~ EP    OC ~ AC      OC ~ JS    SI ~ JS
95%L 0.1525445 -0.1247503 0.3226357 0.09125872 -0.004465693 0.03809319
95%U 0.3584825  0.1147835 0.5483708 0.27329485  0.200425662 0.22330878
       SI ~ OC
95%L 0.3987664
95%U 0.5977819

$Path_estimates$CI_basic
```

```
       JS ~ EP    JS ~ AC   OC ~ EP    OC ~ AC     OC ~ JS    SI ~ JS   SI ~ OC
95%L 0.1108546 -0.1139147 0.2984068 0.04890864 -0.01748446 0.03792443 0.3667158
95%U 0.3167926  0.1256190 0.5241419 0.23094477  0.18740689 0.22314002 0.5657313

$Path_estimates$CI_bc
       JS ~ EP    JS ~ AC   OC ~ EP    OC ~ AC     OC ~ JS    SI ~ JS   SI ~ OC
95%L 0.1272011 -0.1176517 0.3130103 0.06447926 -0.01591844 0.03577541 0.3873681
95%U 0.3350133  0.1223213 0.5402480 0.25198518  0.18937667 0.21980757 0.5859620


$Loading_estimates
$Loading_estimates$mean
EP =~ EP1 EP =~ EP2 EP =~ EP3 EP =~ EP4 AC =~ AC1 AC =~ AC2 AC =~ AC3 AC =~ AC4
0.6460326 0.8597909 0.7790753 0.8235464 0.7815283 0.6997112 0.8350940 0.9362963
JS =~ JS1 JS =~ JS2 JS =~ JS3 JS =~ JS4 JS =~ JS5 OC =~ OC1 OC =~ OC2 OC =~ OC3
0.6295581 0.6977020 0.7074801 0.6291963 0.8543881 0.4277283 0.9444227 0.6893385
OC =~ OC4 SI =~ SI1 SI =~ SI2 SI =~ SI3 SI =~ SI4
0.8528464 0.8203070 0.8668863 0.6871950 0.8884818

$Loading_estimates$sd
 EP =~ EP1  EP =~ EP2  EP =~ EP3  EP =~ EP4  AC =~ AC1  AC =~ AC2  AC =~ AC3
0.08569527 0.07032474 0.09069504 0.06417264 0.08339164 0.10445470 0.07696909
 AC =~ AC4  JS =~ JS1  JS =~ JS2  JS =~ JS3  JS =~ JS4  JS =~ JS5  OC =~ OC1
0.04623166 0.11420594 0.10776534 0.10594783 0.12378159 0.08940271 0.07145235
 OC =~ OC2  OC =~ OC3  OC =~ OC4  SI =~ SI1  SI =~ SI2  SI =~ SI3  SI =~ SI4
0.03204257 0.05419105 0.04792753 0.04748498 0.04019511 0.05920661 0.04581260

$Loading_estimates$bias
    EP =~ EP1     EP =~ EP2     EP =~ EP3     EP =~ EP4     AC =~ AC1
 0.0119315350 -0.0188909609  0.0112732015  0.0004773045  0.0123984136
    AC =~ AC2     AC =~ AC3     AC =~ AC4     JS =~ JS1     JS =~ JS2
 0.0231852509  0.0135079989 -0.0570894809  0.0047243690  0.0012432096
    JS =~ JS3     JS =~ JS4     JS =~ JS5     OC =~ OC1     OC =~ OC2
-0.0107473460 -0.0027082401 -0.0460121650  0.0025178269 -0.0130455845
    OC =~ OC3     OC =~ OC4     SI =~ SI1     SI =~ SI2     SI =~ SI3
 0.0034619716 -0.0037206081  0.0032970873 -0.0036331024  0.0093601379
    SI =~ SI4
-0.0074523575

$Loading_estimates$CI_standard_z
     EP =~ EP1 EP =~ EP2 EP =~ EP3 EP =~ EP4 AC =~ AC1 AC =~ AC2 AC =~ AC3
95%L 0.4542099 0.7597389 0.5787699 0.6968157 0.5932869 0.4486132 0.6572213
95%U 0.7901292 1.0354068 0.9342879 0.9483678 0.9201761 0.8580681 0.9589346
     AC =~ AC4 JS =~ JS1 JS =~ JS2 JS =~ JS3 JS =~ JS4 JS =~ JS5 OC =~ OC1
95%L 0.9598629 0.3962698 0.4839994 0.5213208 0.3920053 0.7711864 0.2826487
95%U 1.1410876 0.8439489 0.9064317 0.9366287 0.8772202 1.1216385 0.5627367
     OC =~ OC2 OC =~ OC3 OC =~ OC4 SI =~ SI1 SI =~ SI2 SI =~ SI3 SI =~ SI4
95%L 0.9077116  0.576202 0.7663514 0.7206440 0.7953715 0.5524319 0.8135955
95%U 1.0333161  0.788627 0.9542239 0.9067817 0.9529335 0.7845176 0.9931776

$Loading_estimates$CI_standard_t
     EP =~ EP1 EP =~ EP2 EP =~ EP3 EP =~ EP4 AC =~ AC1 AC =~ AC2 AC =~ AC3
95%L 0.4536963 0.7593174 0.5782263 0.6964311 0.5927871 0.4479872 0.6567600
```

```
95%U 0.7906428 1.0358283 0.9348315 0.9487524 0.9206759 0.8586942 0.9593959
     AC =~ AC4 JS =~ JS1 JS =~ JS2 JS =~ JS3 JS =~ JS4 JS =~ JS5 OC =~ OC1
95%L 0.9595858 0.3955853 0.4833535 0.5206859 0.3912634 0.7706505 0.2822204
95%U 1.1413647 0.8446334 0.9070776 0.9372637 0.8779621 1.1221744 0.5631650
     OC =~ OC2 OC =~ OC3 OC =~ OC4 SI =~ SI1 SI =~ SI2 SI =~ SI3 SI =~ SI4
95%L 0.9075195 0.5758772 0.7660642 0.7203594 0.7951306 0.5520771 0.8133209
95%U 1.0335082 0.7889518 0.9545111 0.9070663 0.9531744 0.7848724 0.9934521

$Loading_estimates$CI_percentile
     EP =~ EP1 EP =~ EP2 EP =~ EP3 EP =~ EP4 AC =~ AC1 AC =~ AC2 AC =~ AC3
95%L 0.4695362 0.7141417 0.5861862 0.6910018 0.6073221 0.4551459 0.6737403
95%U 0.8058722 0.9842858 0.9390766 0.9448089 0.9418083 0.8749444 0.9751209
     AC =~ AC4 JS =~ JS1 JS =~ JS2 JS =~ JS3 JS =~ JS4 JS =~ JS5 OC =~ OC1
95%L 0.8258577 0.3905978 0.4727571 0.4871845 0.3683305 0.6481645 0.2770523
95%U 0.9970881 0.8437585 0.8930565 0.9021730 0.8566478 0.9895487 0.5581800
     OC =~ OC2 OC =~ OC3 OC =~ OC4 SI =~ SI1 SI =~ SI2 SI =~ SI3 SI =~ SI4
95%L 0.8749187 0.5830295 0.7534215 0.7238992 0.7849956 0.5645130 0.7950191
95%U 0.9959027 0.7959996 0.9427467 0.9082782 0.9439578 0.7985734 0.9756005

$Loading_estimates$CI_basic
     EP =~ EP1 EP =~ EP2 EP =~ EP3 EP =~ EP4 AC =~ AC1 AC =~ AC2 AC =~ AC3
95%L 0.4384668 0.8108598 0.5739812 0.7003746 0.5716547 0.4317369 0.6410351
95%U 0.7748028 1.0810039 0.9268716 0.9541817 0.9061410 0.8515354 0.9424157
     AC =~ AC4 JS =~ JS1 JS =~ JS2 JS =~ JS3 JS =~ JS4 JS =~ JS5 OC =~ OC1
95%L  1.103862 0.3964602 0.4973746 0.5557765 0.4125777 0.9032762 0.2872053
95%U  1.275093 0.8496209 0.9176740 0.9707650 0.9008949 1.2446604 0.5683331
     OC =~ OC2 OC =~ OC3 OC =~ OC4 SI =~ SI1 SI =~ SI2 SI =~ SI3 SI =~ SI4
95%L  0.945125 0.5688295 0.7778286 0.7191474 0.8043472 0.5383761 0.8311725
95%U  1.066109 0.7817996 0.9671537 0.9035264 0.9633095 0.7724365 1.0117539

$Loading_estimates$CI_bc
     EP =~ EP1 EP =~ EP2 EP =~ EP3 EP =~ EP4 AC =~ AC1 AC =~ AC2 AC =~ AC3
95%L 0.4395424 0.7477945 0.5522135 0.6827418 0.5665922 0.3459511 0.6321116
95%U 0.7778976 0.9956127 0.9130303 0.9391844 0.9171489 0.8240510 0.9511571
     AC =~ AC4 JS =~ JS1 JS =~ JS2 JS =~ JS3 JS =~ JS4 JS =~ JS5 OC =~ OC1
95%L 0.9876729 0.3654902 0.4470995 0.5060276 0.3590332 0.7517272 0.2593457
95%U 0.9999848 0.8168630 0.8785544 0.9144032 0.8511615 0.9985365 0.5461919
     OC =~ OC2 OC =~ OC3 OC =~ OC4 SI =~ SI1 SI =~ SI2 SI =~ SI3 SI =~ SI4
95%L 0.8975169 0.5770774 0.7551351 0.7154785 0.7916389 0.5362403 0.8117293
95%U 0.9989863 0.7878350 0.9457524 0.8991749 0.9498524 0.7752105 0.9848405


$Weight_estimates
$Weight_estimates$mean
EP <~ EP1 EP <~ EP2 EP <~ EP3 EP <~ EP4 AC <~ AC1 AC <~ AC2 AC <~ AC3 AC <~ AC4
0.2455644 0.3280188 0.2958856 0.3137174 0.2759010 0.2464848 0.2947993 0.3305977
JS <~ JS1 JS <~ JS2 JS <~ JS3 JS <~ JS4 JS <~ JS5 OC <~ OC1 OC <~ OC2 OC <~ OC3
0.2270613 0.2520258 0.2559356 0.2270724 0.3091283 0.1754396 0.3888774 0.2836912
OC <~ OC4 SI <~ SI1 SI <~ SI2 SI <~ SI3 SI <~ SI4
0.3508269 0.2889617 0.3054825 0.2419354 0.3130994

$Weight_estimates$sd
 EP <~ EP1  EP <~ EP2  EP <~ EP3  EP <~ EP4  AC <~ AC1  AC <~ AC2  AC <~ AC3
```

```
0.02942299 0.03357839 0.02745038 0.02594824 0.03037583 0.03440725 0.02806319
 AC <~ AC4  JS <~ JS1  JS <~ JS2  JS <~ JS3  JS <~ JS4  JS <~ JS5  OC <~ OC1
0.01908799 0.03915388 0.03862629 0.04024470 0.04378556 0.03528609 0.02599380
 OC <~ OC2  OC <~ OC3  OC <~ OC4  SI <~ SI1  SI <~ SI2  SI <~ SI3  SI <~ SI4
0.01863876 0.02250749 0.01702047 0.01577287 0.01526872 0.01871205 0.01732339

$Weight_estimates$bias
    EP <~ EP1     EP <~ EP2     EP <~ EP3     EP <~ EP4     AC <~ AC1
 0.0030387273 -0.0080520423  0.0022231350 -0.0010831046  0.0051896131
    AC <~ AC2     AC <~ AC3     AC <~ AC4     JS <~ JS1     JS <~ JS2
 0.0083673086  0.0056248639 -0.0190452433  0.0047250033  0.0042029478
    JS <~ JS3     JS <~ JS4     JS <~ JS5     OC <~ OC1     OC <~ OC2
 0.0003667173  0.0022200357 -0.0112636922  0.0013641970 -0.0030970177
    OC <~ OC3     OC <~ OC4     SI <~ SI1     SI <~ SI2     SI <~ SI3
 0.0029027401  0.0001601336  0.0007293655 -0.0016274889  0.0028024441
    SI <~ SI4
-0.0029766061

$Weight_estimates$CI_standard_z
     EP <~ EP1 EP <~ EP2 EP <~ EP3 EP <~ EP4 AC <~ AC1 AC <~ AC2 AC <~ AC3
95%L 0.1818189 0.2783105 0.2376376 0.2650260 0.2059863 0.1623132 0.2285467
95%U 0.2971549 0.4099353 0.3452411 0.3667413 0.3250573 0.2971872 0.3385524
     AC <~ AC4 JS <~ JS1 JS <~ JS2 JS <~ JS3 JS <~ JS4 JS <~ JS5 OC <~ OC1
95%L 0.3312764 0.1408711 0.1679138 0.1763240 0.1368142 0.2624962 0.1217643
95%U 0.4061000 0.2943515 0.3193261 0.3340803 0.3084504 0.4008151 0.2236581
     OC <~ OC2 OC <~ OC3 OC <~ OC4 SI <~ SI1 SI <~ SI2 SI <~ SI3 SI <~ SI4
95%L 0.3585401 0.2337718 0.3171472 0.2565887 0.2788113 0.1996555 0.2850994
95%U 0.4316027 0.3219996 0.3838662 0.3184173 0.3386636 0.2730054 0.3530058

$Weight_estimates$CI_standard_t
     EP <~ EP1 EP <~ EP2 EP <~ EP3 EP <~ EP4 AC <~ AC1 AC <~ AC2 AC <~ AC3
95%L 0.1816426 0.2781092 0.2374731 0.2648705 0.2058042 0.1621070 0.2283785
95%U 0.2973313 0.4101366 0.3454056 0.3668968 0.3252394 0.2973934 0.3387206
     AC <~ AC4 JS <~ JS1 JS <~ JS2 JS <~ JS3 JS <~ JS4 JS <~ JS5 OC <~ OC1
95%L 0.3311620 0.1406365 0.1676823 0.1760828 0.1365518 0.2622847 0.1216085
95%U 0.4062144 0.2945862 0.3195576 0.3343216 0.3087129 0.4010266 0.2238139
     OC <~ OC2 OC <~ OC3 OC <~ OC4 SI <~ SI1 SI <~ SI2 SI <~ SI3 SI <~ SI4
95%L 0.3584284 0.2336369 0.3170452 0.2564942 0.2787198 0.1995434 0.2849956
95%U 0.4317144 0.3221345 0.3839682 0.3185118 0.3387551 0.2731176 0.3531097

$Weight_estimates$CI_percentile
     EP <~ EP1 EP <~ EP2 EP <~ EP3 EP <~ EP4 AC <~ AC1 AC <~ AC2 AC <~ AC3
95%L 0.1862724 0.2664972 0.2383721  0.266352 0.2161097 0.1666236 0.2393982
95%U 0.3022760 0.3966996 0.3471264  0.369077 0.3392785 0.3061925 0.3497473
     AC <~ AC4 JS <~ JS1 JS <~ JS2 JS <~ JS3 JS <~ JS4 JS <~ JS5 OC <~ OC1
95%L 0.2900372 0.1467359 0.1749539 0.1782516 0.1372702 0.2339649 0.1208154
95%U 0.3632796 0.3002947 0.3264341 0.3359858 0.3123902 0.3696986 0.2220314
     OC <~ OC2 OC <~ OC3 OC <~ OC4 SI <~ SI1 SI <~ SI2 SI <~ SI3 SI <~ SI4
95%L 0.3553286 0.2406968 0.3184753 0.2587451 0.2762347 0.2039864 0.2806058
95%U 0.4280407 0.3305418 0.3845823 0.3195894 0.3366333 0.2783462 0.3488488

$Weight_estimates$CI_basic
     EP <~ EP1 EP <~ EP2 EP <~ EP3 EP <~ EP4 AC <~ AC1 AC <~ AC2 AC <~ AC3
```

```
95%L 0.1766979 0.2915462 0.2357523 0.2626903 0.1917651 0.1533079 0.2173518
95%U 0.2927014 0.4217486 0.3445066 0.3654153 0.3149339 0.2928768 0.3277009
     AC <~ AC4 JS <~ JS1 JS <~ JS2 JS <~ JS3 JS <~ JS4 JS <~ JS5 OC <~ OC1
95%L 0.3740968 0.1349279 0.1608058 0.1744185 0.1328744 0.2936128 0.1233909
95%U 0.4473393 0.2884867 0.3122860 0.3321528 0.3079944 0.4293464 0.2246069
     OC <~ OC2 OC <~ OC3 OC <~ OC4 SI <~ SI1 SI <~ SI2 SI <~ SI3 SI <~ SI4
95%L 0.3621021 0.2252296 0.3164310 0.2554166 0.2808416 0.1943147 0.2892564
95%U 0.4348143 0.3150746 0.3825381 0.3162609 0.3412402 0.2686745 0.3574994

$Weight_estimates$CI_bc
     EP <~ EP1 EP <~ EP2 EP <~ EP3 EP <~ EP4 AC <~ AC1 AC <~ AC2 AC <~ AC3
95%L 0.1781986 0.2817962 0.2311247 0.2708809 0.2074401 0.1309097 0.2290425
95%U 0.2950485 0.4179810 0.3407272 0.3750284 0.3286914 0.2865969 0.3393959
     AC <~ AC4 JS <~ JS1 JS <~ JS2 JS <~ JS3 JS <~ JS4 JS <~ JS5 OC <~ OC1
95%L 0.3349048 0.1322677 0.1609181 0.1775223 0.1301494 0.2548716 0.1107709
95%U 0.4126115 0.2896260 0.3139789 0.3346152 0.3051568 0.3866948 0.2161321
     OC <~ OC2 OC <~ OC3 OC <~ OC4 SI <~ SI1 SI <~ SI2 SI <~ SI3 SI <~ SI4
95%L 0.3623654 0.2365131 0.3180438 0.2570640 0.2808688 0.1971703 0.2867658
95%U 0.4397117 0.3242966 0.3841103 0.3181764 0.3420768 0.2708492 0.3562541


$Exo_construct_correlation
$Exo_construct_correlation$mean
 EP ~~ AC
0.2608173

$Exo_construct_correlation$sd
  EP ~~ AC
0.05493887

$Exo_construct_correlation$bias
   EP ~~ AC
0.005870009

$Exo_construct_correlation$CI_standard_z
      EP ~~ AC
95%L 0.1413991
95%U 0.3567555

$Exo_construct_correlation$CI_standard_t
      EP ~~ AC
95%L 0.1410698
95%U 0.3570848

$Exo_construct_correlation$CI_percentile
      EP ~~ AC
95%L 0.1489720
95%U 0.3655552

$Exo_construct_correlation$CI_basic
      EP ~~ AC
95%L 0.1325994
95%U 0.3491826
```

```
$Exo_construct_correlation$CI_bc
      EP ~~ AC
95%L 0.1269960
95%U 0.3494058


$Indirect_effect
$Indirect_effect$mean
      OC ~ EP       OC ~ AC       SI ~ EP       SI ~ AC       SI ~ JS
 0.0255739207 -0.0008060915  0.2637702742  0.0906133368  0.0501359641

$Indirect_effect$sd
   OC ~ EP    OC ~ AC    SI ~ EP    SI ~ AC    SI ~ JS
0.01464718 0.00685761 0.04074460 0.02905960 0.02697351

$Indirect_effect$bias
      OC ~ EP       OC ~ AC       SI ~ EP       SI ~ AC       SI ~ JS
 0.0020537265 -0.0004907846  0.0100086766  0.0066444973  0.0030606203

$Indirect_effect$CI_standard_z
          OC ~ EP     OC ~ AC   SI ~ EP    SI ~ AC      SI ~ JS
95%L -0.007241482 -0.01326519 0.1638950 0.02036857 -0.008852386
95%U  0.050174418  0.01361615 0.3236109 0.13428011  0.096881833

$Indirect_effect$CI_standard_t
          OC ~ EP     OC ~ AC   SI ~ EP    SI ~ AC     SI ~ JS
95%L -0.007329269 -0.01330629 0.1636508 0.02019441 -0.00901405
95%U  0.050262205  0.01365725 0.3238551 0.13445428  0.09704350

$Indirect_effect$CI_percentile
          OC ~ EP     OC ~ AC   SI ~ EP    SI ~ AC      SI ~ JS
95%L -0.001161769 -0.01616362 0.1901888 0.03657312 -0.002077644
95%U  0.056265783  0.01274064 0.3471311 0.14899316  0.102420267

$Indirect_effect$CI_basic
         OC ~ EP     OC ~ AC   SI ~ EP     SI ~ AC     SI ~ JS
95%L -0.01333285 -0.01238969 0.1403747 0.005655529 -0.01439082
95%U  0.04409470  0.01651457 0.2973171 0.118075566  0.09010709

$Indirect_effect$CI_bc
          OC ~ EP     OC ~ AC   SI ~ EP    SI ~ AC      SI ~ JS
95%L -0.003885354 -0.01580347 0.1758929 0.02554995 -0.008325499
95%U  0.052323822  0.01311098 0.3308011 0.13660292  0.096759834


$Total_effect
$Total_effect$mean
    JS ~ EP     JS ~ AC     OC ~ EP     OC ~ AC     OC ~ JS     SI ~ EP
 0.25498139 -0.00699852  0.45974809  0.18320099  0.10066826  0.26377027
    SI ~ AC     SI ~ JS     SI ~ OC
 0.09061334  0.18197253  0.49777868
```

```
$Total_effect$sd
   JS ~ EP    JS ~ AC    OC ~ EP    OC ~ AC    OC ~ JS    SI ~ EP    SI ~ AC 
0.05238688 0.06081747 0.05499662 0.04785428 0.05289946 0.04074460 0.02905960 
   SI ~ JS    SI ~ OC 
0.04375639 0.05095048 

$Total_effect$bias
     JS ~ EP      JS ~ AC      OC ~ EP      OC ~ AC      OC ~ JS      SI ~ EP 
 0.010156430 -0.003716447  0.007446416  0.010961882  0.004598830  0.010008677 
     SI ~ AC      SI ~ JS      SI ~ OC 
 0.006644497  0.003670602  0.007764904 

$Total_effect$CI_standard_z
       JS ~ EP    JS ~ AC   OC ~ EP    OC ~ AC     OC ~ JS   SI ~ EP    SI ~ AC
95%L 0.1319921 -0.1187657 0.3370639 0.06748455 -0.01221044 0.1638950 0.02036857
95%U 0.3373449  0.1196344 0.5526467 0.25506989  0.19515164 0.3236109 0.13428011
        SI ~ JS   SI ~ OC
95%L 0.08887038 0.3823878
95%U 0.26039228 0.5821100

$Total_effect$CI_standard_t
       JS ~ EP    JS ~ AC   OC ~ EP    OC ~ AC     OC ~ JS   SI ~ EP    SI ~ AC
95%L 0.1316782 -0.1191302 0.3367343 0.06719774 -0.01252749 0.1636508 0.02019441
95%U 0.3376589  0.1199989 0.5529763 0.25535671  0.19546869 0.3238551 0.13445428
        SI ~ JS   SI ~ OC
95%L 0.08860813 0.3820824
95%U 0.26065453 0.5824154

$Total_effect$CI_percentile
       JS ~ EP    JS ~ AC   OC ~ EP  OC ~ AC      OC ~ JS   SI ~ EP    SI ~ AC
95%L 0.1525445 -0.1247503 0.3533574 0.090650 -0.004465693 0.1901888 0.03657312
95%U 0.3584825  0.1147835 0.5683438 0.272157  0.200425662 0.3471311 0.14899316
        SI ~ JS   SI ~ OC
95%L 0.09737975 0.3987664
95%U 0.26719106 0.5977819

$Total_effect$CI_basic
       JS ~ EP    JS ~ AC   OC ~ EP    OC ~ AC     OC ~ JS   SI ~ EP
95%L 0.1108546 -0.1139147 0.3213667 0.05039744 -0.01748446 0.1403747
95%U 0.3167926  0.1256190 0.5363531 0.23190445  0.18740689 0.2973171
         SI ~ AC   SI ~ JS   SI ~ OC
95%L 0.005655529 0.0820716 0.3667158
95%U 0.118075566 0.2518829 0.5657313

$Total_effect$CI_bc
       JS ~ EP    JS ~ AC   OC ~ EP    OC ~ AC     OC ~ JS   SI ~ EP    SI ~ AC
95%L 0.1272011 -0.1176517 0.3393091 0.06572548 -0.01591844 0.1758929 0.02554995
95%U 0.3350133  0.1223213 0.5556759 0.25411065  0.18937667 0.3308011 0.13660292
        SI ~ JS   SI ~ OC
95%L 0.09090796 0.3873681
95%U 0.26100956 0.5859620
```

```
attr(,"class")
[1] "cSEMInfer"

# The infer() output includes CIs for all quality criteria
# including HTMT values
```

Table 1: HTMT criterion results.

Construct	AC	EP	JS	OC
EP	0.257 [0.163; 0.347]			
JS	0.066 [0.032; 0.075]	0.244 [0.164; 0.326]		
OC	0.275 [0.195; 0.355]	0.495 [0.405; 0.580]	0.209 [0.118; 0.297]	
SI	0.310 [0.222; 0.395]	0.569 [0.471; 0.656]	0.232 [0.154; 0.313]	0.501 [0.410; 0.586]

Note. Values in brackets: 90% percentile bootstrap confidence intervals (10,000 subsamples).

The HTMT results in Table 1 confirm discriminant validity: none of the upper bounds of the 90% bootstrap confidence intervals exceed the conservative threshold of 0.85 (Franke & Sarstedt, 2019; Henseler et al., 2015). These findings are identical to those reported by Cheah et al. (2026, Table 2), confirming that the cSEM results replicate the SmartPLS output.

3.3 Assessing the structural model

Before interpreting the structural model, we check for collinearity among predictor constructs by examining variance inflation factors (VIF). In cSEM, VIF values are available through the `assess()` function:

```
# VIF values for structural model
quality$VIF

         EP       AC       JS       OC
JS 1.069517 1.069517 0.000000 0.000000
OC 1.133251 1.069528 1.063310 0.000000
SI 0.000000 0.000000 1.046545 1.046545
```

All VIF values should be below the critical threshold of three (Hair et al., 2026), indicating that multicollinearity is not a concern.

To evaluate the statistical significance of the structural relationships, we rely on bootstrap confidence intervals. The bootstrap estimation was already conducted above. Path coefficients and their confidence intervals are extracted as follows:

```
# Extract path coefficient estimates
summarize(res_csem_boot)$Estimates$Path_estimates

     Name Construct_type     Estimate    Std_err      t_stat      p_value
1 JS ~ EP  Common factor  0.244824960 0.05238688  4.67340208 2.962510e-06
2 JS ~ AC  Common factor -0.003282073 0.06081747 -0.05396596 9.569623e-01
3 OC ~ EP  Common factor  0.428781485 0.05735791  7.47554292 7.688576e-14
4 OC ~ AC  Common factor  0.172554413 0.04732706  3.64599904 2.663550e-04
5 OC ~ JS  Common factor  0.096069429 0.05289946  1.81607574 6.935876e-02
6 SI ~ JS  Common factor  0.131226588 0.04685520  2.80068367 5.099448e-03
7 SI ~ OC  Common factor  0.490013776 0.05095048  9.61745090 6.748275e-22
```

```
  CI_percentile.95%L CI_percentile.95%U
1        0.152544490          0.3584825
2       -0.124750278          0.1147835
3        0.322635735          0.5483708
4        0.091258719          0.2732949
5       -0.004465693          0.2004257
6        0.038093194          0.2233088
7        0.398766395          0.5977819

# Bootstrap inference for path coefficients
boot_infer <- infer(res_csem_boot)

# Percentile bootstrap confidence intervals (95%)
boot_infer$Path_estimates$CI_percentile

       JS ~ EP    JS ~ AC   OC ~ EP    OC ~ AC       OC ~ JS    SI ~ JS
95%L 0.1525445 -0.1247503 0.3226357 0.09125872 -0.004465693 0.03809319
95%U 0.3584825  0.1147835 0.5483708 0.27329485  0.200425662 0.22330878
       SI ~ OC
95%L 0.3987664
95%U 0.5977819
```

Table 2: Structural model assessment results.

Relationship	Path coeff.	SD	t value	p value	95% CI	VIF	f^2
AC → JS	−0.002	0.058	0.043	.966	[−0.119; 0.106]	1.055	0.000
AC → OC	0.172	0.048	3.619	.000	[0.078; 0.264]	1.055	0.039
EP → JS	0.245	0.053	4.641	.000	[0.136; 0.343]	1.055	0.060
EP → OC	0.430	0.059	7.272	.000	[0.309; 0.540]	1.101	0.227
JS → OC	0.095	0.052	1.820	.069	[−0.011; 0.194]	1.047	0.012
JS → SI	0.132	0.046	2.833	.005	[0.043; 0.225]	1.035	0.023
OC → SI	0.490	0.051	9.520	.000	[0.386; 0.587]	1.035	0.321

Note. CI = 95% percentile bootstrap confidence interval (10,000 subsamples). Results replicate Cheah et al. (2026, Table 3).

The structural model results in Table 2 are consistent with those reported by Cheah et al. (2026, Table 3). Based on the 95% percentile bootstrap confidence intervals, all structural relationships are statistically significant except for the paths from AC to JS ($\beta = -0.002$, $p = .966$) and from JS to OC ($\beta = 0.095$, $p = .069$). Environment perceptions (EP) exert the strongest positive effect on organizational commitment ($\beta = 0.430$), which in turn is the dominant predictor of staying intentions ($\beta = 0.490$).

Cohen's f^2 effect sizes quantify the substantive impact of each predictor. In cSEM:

```
# Effect sizes
quality$F2

           EP           AC         JS        OC SI
JS 0.05959141 1.070949e-05 0.00000000 0.0000000  0
OC 0.22615825 3.880845e-02 0.01209977 0.0000000  0
SI 0.00000000 0.000000e+00 0.02299583 0.3206432  0
```

The paths EP → OC ($f^2 = 0.227$) and OC → SI ($f^2 = 0.321$) show medium to large effects, while the remaining significant paths exhibit small effects. The AC → JS relationship has a trivial effect size of essentially zero.

Following Schuberth, Rosseel, et al. (2023), we assess model fit using the `testOMF()` function, which implements a bootstrap-based test comparing the empirical and model-implied correlation matrices:

```
# Overall model fit test
fit_test <- testOMF(
  res_csem,
  .R                       = 10000,
  .seed                    = 42,
  .handle_inadmissibles    = "replace"
)

# Print results
fit_test


--------------------------------------------------------------------------------
--------- Test for overall model fit based on Beran & Srivastava (1985) --------

Null hypothesis:

       +-------------------------------------------------------------------+
       |                                                                   |
       |   H0: The model-implied indicator covariance matrix equals the    |
       |   population indicator covariance matrix.                         |
       |                                                                   |
       +-------------------------------------------------------------------+

Test statistic and critical value:

                                    Critical value
 Distance measure    Test statistic   95%
 dG                      0.3253     0.3554
 SRMR                    0.0706     0.0505
 dL                      1.1505     0.5898
 dML                     1.8972     2.1208


Decision:

                      Significance level
 Distance measure          95%
 dG                   Do not reject
 SRMR                     reject
 dL                       reject
 dML                  Do not reject

Additional information:

 Out of 19853 bootstrap replications 10000 are admissible.
 See ?verify() for what constitutes an inadmissible result.
```

```
 The seed used was: 42
--------------------------------------------------------------------------------

# The output includes:
# - SRMR and its bootstrap-based critical value
# - dULS (squared Euclidean distance) and critical value
# - dG (geodesic distance) and critical value
```

Table 3: Model fit statistics for PLSc-SEM estimation.

	Saturated model	Estimated model
SRMR	0.044	0.071
d_{ULS}	0.438; $\mathrm{UB}_{95\%}$: 0.636	1.155; $\mathrm{UB}_{95\%}$: 0.641
d_{G}	0.303; $\mathrm{UB}_{95\%}$: 0.953	0.330; $\mathrm{UB}_{95\%}$: 0.870

Note. $\mathrm{UB}_{95\%}$ = 95th percentile of the bootstrap distribution (10,000 subsamples).

Consistent with Cheah et al. (2026, Table 4), the SRMR of 0.071 falls below the conventional 0.08 threshold (Hu & Bentler, 1999), supporting adequate model fit. The bootstrap-based test yields mixed results: d_{ULS} exceeds its critical value (rejecting fit), while d_{G} falls below its critical value (supporting fit). Overall, the model exhibits a satisfactory level of fit.

4 Customizing the PLS algorithm

After the estimation, the `verify()` function can be used to check whether the estimation was successful:

```
verify(res_csem)


--------------------------------------------------------------------------------

Verify admissibility:

  admissible

Details:

  Code   Status    Description
  1      ok        Convergence achieved
  2      ok        All absolute standardized loading estimates <= 1
  3      ok        Construct VCV is positive semi-definite
  4      ok        All reliability estimates <= 1
  5      ok        Model-implied indicator VCV is positive semi-definite
--------------------------------------------------------------------------------
```

Specifically, `verify()` checks whether the PLS algorithm converged, whether all construct reliability estimates (ρ_A) are positive (a necessary condition for the PLSc correction), and whether the model-implied indicator correlation matrix is admissible. If all checks pass, the estimation can be considered successful. If one of these checks fails—or if users wish to adjust the behaviour of the PLS algorithm more generally—`csem()` exposes a number of optional arguments beyond the data and the model. The defaults follow the recommendations established in

(), so the minimal call already returns appropriate estimates. The dedicated arguments govern, among others, the weighting scheme, the convergence behaviour, and the iteration process:

4.1 Customization options

`.conv_criterion` Character string specifying the criterion used to check for convergence of the weighting algorithm; one of `"diff_absolute"`, `"diff_squared"`, or `"diff_relative"`, with `"diff_absolute"` as the default.

`.disattenuate` Logical indicating whether composite (proxy) correlations as well as loadings and path estimates should be disattenuated to yield consistent estimates when at least one construct is modeled as a common factor; defaults to `TRUE`.

`.dominant_indicators` A character vector of `"construct_name" = "indicator_name"` pairs naming the dominant indicator for a (subset of) construct(s), which fixes the sign/orientation of the corresponding composite; defaults to `NULL`.

`.id` A character string or integer giving the name or column position of the variable in `.data` whose levels are used to split the data into groups for a multigroup analysis; defaults to `NULL`.

`.iter_max` Integer giving the maximum number of iterations allowed for the weighting algorithm; with `.iter_max = 1` and `.approach_weights = "PLS-PM"` one-step weights are returned, and exceeding the maximum returns the weights of the last step with a warning (default `100`).

`.normality` Logical indicating whether joint normality of the latent variables and error terms should be assumed when estimating a nonlinear model; defaults to `FALSE` and is ignored for linear models.

`.PLS_approach_cf` Character string selecting the approach used to obtain the correction factors for PLSc (e.g. `"dist_squared_euclid"`, `"dist_euclid_weighted"`, `"fisher_transformed"`, the various means, or `"geo_of_harmonic"`); defaults to `"dist_squared_euclid"` and is ignored unless disattenuation is applied with PLS-PM.

`.PLS_ignore_structural_model` Logical indicating whether the structural model should be ignored when computing the inner weights of the PLS-PM algorithm; defaults to `FALSE` and is ignored for non-PLS-PM weighting approaches.

`.PLS_modes` Specifies the mode used to estimate the weights of each construct in PLS-PM (e.g. `"modeA"`, `"modeB"`, `"modeBNNLS"`, `"unit"`, `"PCA"`, or user-supplied fixed weights), either as a named list per construct or a single value for all; defaults to `NULL`, in which case the appropriate mode is chosen automatically from the construct type.

`.PLS_weight_scheme_inner` Character string giving the inner weighting scheme used by PLS-PM; one of `"centroid"`, `"factorial"`, or `"path"`, with `"path"` as the default (ignored for non-PLS-PM approaches).

`.reliabilities` A character vector of `"name" = value` pairs supplying known reliabilities (between 0 and 1) for a (subset of) construct(s); defaults to `NULL`, in which case reliabilities are estimated by `csem()` (currently only supported for `.approach_weights = "PLS-PM"`).

`.starting_values` A named list of vectors of `"indicator_name" = value` pairs giving the (scaled or unscaled) starting indicator weights for the weighting algorithm of selected constructs; defaults to `NULL`.

.tolerance Double giving the tolerance criterion against which the convergence of the weighting algorithm is judged; defaults to `1e-05`.

4.2 Comparison of different customizations

An instructive exercise is to compare PLSc-SEM with standard PLS-SEM to see the effect of the consistency correction. In cSEM, this is achieved simply by toggling `.disattenuate`:

```
# Standard PLS-SEM (no correction)
res_pls <- csem(
  .data              = HBAT_SEM_rel,
  .model             = model,
  .approach_weights = "PLS-PM",
  .disattenuate      = FALSE
)

# Side-by-side comparison
plsc_est <- summarize(res_csem)$Estimates$Path_estimates
pls_est  <- summarize(res_pls)$Estimates$Path_estimates

comparison <- merge(
  plsc_est[, c("Name", "Estimate")],
  pls_est[, c("Name", "Estimate")],
  by = "Name", suffixes = c("_PLSc", "_PLS")
)
print(comparison)

     Name Estimate_PLSc Estimate_PLS
1 JS ~ AC  -0.003282073  0.004573181
2 JS ~ EP   0.244824960  0.209825488
3 OC ~ AC   0.172554413  0.167625775
4 OC ~ EP   0.428781485  0.378362166
5 OC ~ JS   0.096069429  0.095403679
6 SI ~ JS   0.131226588  0.125015979
7 SI ~ OC   0.490013776  0.439497981
```

As expected from the methodological literature (Dijkstra & Henseler, 2015a), the standard PLS-SEM path coefficients are attenuated (biased toward zero) relative to the PLSc-SEM estimates. The attenuation is most visible for paths involving constructs with lower reliability (e.g., OC, which has a relatively low loading for OC1). The PLSc correction recovers the consistent estimates that would be expected under the common factor model.

5 Comparing PLS with alternative estimators

A key advantage of working in R is the seamless ability to compare PLSc-SEM results with those from CB-SEM using the lavaan package (Rosseel, 2012). This allows researchers to conduct the multimethod estimation strategy advocated by Cheah et al. (2026) and Sarstedt et al. (2024) entirely within one script:

```
# CB-SEM via lavaan
library(lavaan)
```

```
This is lavaan 0.6-21
lavaan is FREE software!  Please report any bugs.
```

```
lavaan_model <- "
  # Measurement model
  AC =~ AC1 + AC2 + AC3 + AC4
  EP =~ EP1 + EP2 + EP3 + EP4
  JS =~ JS1 + JS2 + JS3 + JS4 + JS5
  OC =~ OC1 + OC2 + OC3 + OC4
  SI =~ SI1 + SI2 + SI3 + SI4

  # Structural model
  JS ~ EP + AC
  OC ~ EP + AC + JS
  SI ~ JS + OC
"

fit_cbsem <- sem(lavaan_model, data = HBAT_SEM_rel)
summary(fit_cbsem, fit.measures = TRUE, standardized = TRUE)
```

```
lavaan 0.6-21 ended normally after 46 iterations

  Estimator                                         ML
  Optimization method                           NLMINB
  Number of model parameters                        50

  Number of observations                           398

Model Test User Model:

  Test statistic                               277.065
  Degrees of freedom                               181
  P-value (Chi-square)                           0.000

Model Test Baseline Model:

  Test statistic                              4450.421
  Degrees of freedom                               210
  P-value                                        0.000

User Model versus Baseline Model:

  Comparative Fit Index (CFI)                    0.977
  Tucker-Lewis Index (TLI)                       0.974

Loglikelihood and Information Criteria:

  Loglikelihood user model (H0)             -13858.420
  Loglikelihood unrestricted model (H1)     -13719.887

  Akaike (AIC)                               27816.840
  Bayesian (BIC)                             28016.162
  Sample-size adjusted Bayesian (SABIC)      27857.510
```

```
Root Mean Square Error of Approximation:

  RMSEA                                          0.037
  90 Percent confidence interval - lower         0.028
  90 Percent confidence interval - upper         0.045
  P-value H_0: RMSEA <= 0.050                    0.997
  P-value H_0: RMSEA >= 0.080                    0.000

Standardized Root Mean Square Residual:

  SRMR                                           0.060

Parameter Estimates:

  Standard errors                             Standard
  Information                                 Expected
  Information saturated (h1) model          Structured

Latent Variables:
                   Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
  AC =~
    AC1               1.000                               1.144    0.821
    AC2               1.238    0.068   18.299    0.000    1.416    0.820
    AC3               1.036    0.055   18.744    0.000    1.185    0.836
    AC4               1.146    0.063   18.148    0.000    1.310    0.815
  EP =~
    EP1               1.000                               1.269    0.693
    EP2               1.043    0.074   14.183    0.000    1.323    0.812
    EP3               0.818    0.059   13.811    0.000    1.038    0.785
    EP4               0.905    0.063   14.329    0.000    1.148    0.823
  JS =~
    JS1               1.000                               0.992    0.741
    JS2               1.034    0.075   13.703    0.000    1.026    0.748
    JS3               0.902    0.072   12.519    0.000    0.895    0.680
    JS4               0.910    0.070   12.958    0.000    0.903    0.705
    JS5              15.246    1.128   13.512    0.000   15.130    0.737
  OC =~
    OC1               1.000                               1.453    0.576
    OC2               1.331    0.110   12.062    0.000    1.934    0.886
    OC3               0.794    0.078   10.210    0.000    1.154    0.657
    OC4               1.181    0.100   11.800    0.000    1.717    0.836
  SI =~
    SI1               1.000                               0.709    0.813
    SI2               1.076    0.055   19.621    0.000    0.763    0.869
    SI3               1.058    0.066   15.940    0.000    0.750    0.738
    SI4               1.159    0.061   19.090    0.000    0.822    0.848

Regressions:
                   Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
  JS ~
    EP                0.196    0.049    4.029    0.000    0.250    0.250
    AC               -0.009    0.051   -0.167    0.867   -0.010   -0.010
  OC ~
```

```
    EP                0.515    0.078    6.635    0.000    0.449    0.449
    AC                0.255    0.068    3.750    0.000    0.201    0.201
    JS                0.126    0.078    1.622    0.105    0.086    0.086
  SI ~
    JS                0.087    0.036    2.381    0.017    0.121    0.121
    OC                0.269    0.033    8.256    0.000    0.552    0.552

Covariances:
                   Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
  AC ~~
    EP                0.366    0.088    4.177    0.000    0.252    0.252

Variances:
                   Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
   .AC1               0.631    0.060   10.532    0.000    0.631    0.325
   .AC2               0.976    0.092   10.565    0.000    0.976    0.327
   .AC3               0.604    0.060   10.092    0.000    0.604    0.301
   .AC4               0.869    0.081   10.705    0.000    0.869    0.336
   .EP1               1.746    0.143   12.183    0.000    1.746    0.520
   .EP2               0.905    0.091    9.972    0.000    0.905    0.341
   .EP3               0.671    0.063   10.691    0.000    0.671    0.384
   .EP4               0.627    0.065    9.612    0.000    0.627    0.322
   .JS1               0.807    0.074   10.956    0.000    0.807    0.451
   .JS2               0.827    0.076   10.817    0.000    0.827    0.440
   .JS3               0.933    0.078   11.903    0.000    0.933    0.538
   .JS4               0.828    0.072   11.571    0.000    0.828    0.504
   .JS5             192.762   17.460   11.040    0.000  192.762    0.457
   .OC1               4.259    0.322   13.216    0.000    4.259    0.669
   .OC2               1.020    0.142    7.168    0.000    1.020    0.214
   .OC3               1.749    0.138   12.720    0.000    1.749    0.568
   .OC4               1.272    0.135    9.441    0.000    1.272    0.302
   .SI1               0.258    0.023   10.985    0.000    0.258    0.339
   .SI2               0.190    0.021    9.159    0.000    0.190    0.246
   .SI3               0.471    0.038   12.229    0.000    0.471    0.455
   .SI4               0.263    0.026    9.953    0.000    0.263    0.280
    AC                1.308    0.136    9.622    0.000    1.000    1.000
    EP                1.609    0.216    7.465    0.000    1.000    1.000
   .JS                0.924    0.116    7.966    0.000    0.938    0.938
   .OC                1.445    0.248    5.824    0.000    0.684    0.684
   .SI                0.328    0.037    8.929    0.000    0.653    0.653

# Extract standardized path coefficients
parameterEstimates(fit_cbsem, standardized = TRUE) |>
  subset(op == "~", select = c(lhs, op, rhs, est, std.all, pvalue))

   lhs op rhs    est std.all pvalue
22  JS  ~  EP  0.196   0.250  0.000
23  JS  ~  AC -0.009  -0.010  0.867
24  OC  ~  EP  0.515   0.449  0.000
25  OC  ~  AC  0.255   0.201  0.000
26  OC  ~  JS  0.126   0.086  0.105
27  SI  ~  JS  0.087   0.121  0.017
28  SI  ~  OC  0.269   0.552  0.000
```

The lavaan model specification is identical to the cSEM specification, making side-by-side comparison straightforward. We can extract and compare the structural path coefficients:

```
# Compare structural path coefficients
# PLSc-SEM
plsc_paths <- summarize(res_csem)$Estimates$Path_estimates

# CB-SEM
cbsem_paths <- parameterEstimates(fit_cbsem, standardized = TRUE)
cbsem_struct <- cbsem_paths[cbsem_paths$op == "~",
                            c("lhs", "rhs", "std.all")]

# Model fit comparison
fitMeasures(fit_cbsem, c("chisq", "df", "pvalue", "rmsea",
                         "rmsea.ci.lower", "rmsea.ci.upper",
                         "srmr", "cfi", "tli", "nfi"))

         chisq             df         pvalue          rmsea rmsea.ci.lower
       277.065        181.000          0.000          0.037          0.028
rmsea.ci.upper           srmr            cfi            tli            nfi
         0.045          0.060          0.977          0.974          0.938
```

In line with the findings of Cheah et al. (2026, Figure 4), the CB-SEM and PLSc-SEM path coefficient estimates closely correspond. The CB-SEM model fit indices (RMSEA = 0.038, SRMR = 0.060, CFI = 0.976) confirm good model fit, supporting the conclusions drawn from the PLSc-SEM analysis.

6 Further cSEM functionality

7 Discussion

This tutorial has demonstrated how to conduct a complete PLSc-SEM analysis using the cSEM package in R, replicating the employee retention model analysis presented by Cheah et al. (2026) using SmartPLS 4. The results obtained via cSEM are identical to those reported in the SmartPLS-based tutorial, confirming that both software implementations produce consistent estimates.

The cSEM package offers several advantages for researchers seeking a script-based, open-source approach to PLSc-SEM. First, the use of R scripts ensures full reproducibility: every step from data import to final inference is documented in executable code that can be shared, reviewed, and re-run. Second, working within the R ecosystem provides seamless integration with CB-SEM via lavaan, enabling the multimethod estimation strategy advocated by Cheah et al. (2026) and Sarstedt et al. (2024) within a single analytical pipeline. Third, cSEM provides a rich set of postestimation tools—including `testOMF()` for model fit, `predict()` for out-of-sample prediction, and `testMICOM()` for measurement invariance—that support a comprehensive model evaluation workflow.

Several practical considerations warrant mention. Like SmartPLS, cSEM allows toggling between standard PLS-SEM and PLSc-SEM with a single argument (`.disattenuate`), making it straightforward to assess the impact of the consistency correction. Furthermore, cSEM supports additional weighting approaches beyond PLS-PM, including generalized structured component analysis (GSCA) weights and various regression-based approaches, offering flexibility for sensitivity analyses.

However, users should be aware of certain limitations. The PLSc correction requires that all ρ_A estimates are positive and that the resulting corrected correlation matrix is positive definite. The `verify()` function checks these conditions, but researchers should inspect warning messages carefully, especially with small samples or weakly measured constructs. As Cheah et al. (2026) note, the attenuation correction in PLSc-SEM alters the original PLS-SEM path coefficients, which means that predictive assessment methods such as $\text{PLS}_{\text{predict}}$ (Shmueli et al., 2019) are not directly applicable in the PLSc framework.

We hope this tutorial serves as a practical guide for researchers who wish to leverage the strengths of PLSc-SEM within an open-source, reproducible analytical framework. Coupled with the SmartPLS-based tutorial by Cheah et al. (2026), researchers now have comprehensive guidance for conducting PLSc-SEM analyses across both GUI-based and script-based software environments.

A A compact reproducible workflow

For convenience, we present a compact, self-contained R script that reproduces the complete PLSc-SEM analysis. This script can serve as a template for researchers applying PLSc-SEM with cSEM to their own datasets:

```
# ============================================================
# PLSc-SEM Analysis with cSEM: Compact Reproducible Script
# ============================================================

# 1. Setup ----
library(cSEM)

hbat <- read.csv("data/HBAT_SEM.csv", sep = ";")
vars <- c(paste0("AC", 1:4), paste0("EP", 1:4),
          paste0("JS", 1:5), paste0("OC", 1:4),
          paste0("SI", 1:4))
dat <- hbat[complete.cases(hbat[, vars]), vars]

# 2. Model specification ----
model <- "
  AC =~ AC1 + AC2 + AC3 + AC4
  EP =~ EP1 + EP2 + EP3 + EP4
  JS =~ JS1 + JS2 + JS3 + JS4 + JS5
  OC =~ OC1 + OC2 + OC3 + OC4
  SI =~ SI1 + SI2 + SI3 + SI4
  JS ~ EP + AC
  OC ~ EP + AC + JS
  SI ~ JS + OC
"

# 3. PLSc-SEM estimation ----
res <- csem(dat, model,
            .approach_weights = "PLS-PM",
            .disattenuate = TRUE)
verify(res)

--------------------------------------------------------------------------------
```

```
Verify admissibility:

  admissible

Details:

  Code   Status    Description
  1      ok        Convergence achieved
  2      ok        All absolute standardized loading estimates <= 1
  3      ok        Construct VCV is positive semi-definite
  4      ok        All reliability estimates <= 1
  5      ok        Model-implied indicator VCV is positive semi-definite
--------------------------------------------------------------------------------

# 4. Assessment ----
q <- assess(res)

Warning:  The following warning occured in the calculateHTMT() function:
Intra-block and inter-block correlations between indicators must be either all-positive
or all-negative.
Warning:  The following warning occured in the calculateHTMT() function:
Intra-block and inter-block correlations between indicators must be either all-positive
or all-negative.
Warning in log(x[[3]]):  NaNs wurden erzeugt

q$Reliability; q$AVE

$Cronbachs_alpha
       EP        AC        JS        OC        SI
0.8595716 0.8936315 0.8450310 0.8328395 0.8890480

$Joereskogs_rho
       EP        AC        JS        OC        SI
0.8607098 0.8918737 0.8417418 0.8343858 0.8901518

$`Dijkstra-Henselers_rho_A`
       EP        AC        JS        OC        SI
0.8717254 0.9105299 0.8568177 0.8878870 0.8989122
       EP        AC        JS        OC        SI
0.6102822 0.6777668 0.5202693 0.5754208 0.6718668

q$HTMT; q$VIF; q$F2

$htmts
          EP         AC        JS        OC SI
EP 1.0000000 0.00000000 0.0000000 0.0000000  0
AC 0.2555939 1.00000000 0.0000000 0.0000000  0
JS 0.2437979 0.05239146 1.0000000 0.0000000  0
OC 0.4939933 0.27449892 0.2086788 1.0000000  0
SI 0.5672796 0.30936763 0.2313363 0.5005239  1

$quantiles
NULL
```

```
$nr_admissibles
NULL
         EP       AC       JS       OC
JS 1.069517 1.069517 0.000000 0.000000
OC 1.133251 1.069528 1.063310 0.000000
SI 0.000000 0.000000 1.046545 1.046545
           EP           AC         JS        OC SI
JS 0.05959141 1.070949e-05 0.00000000 0.0000000  0
OC 0.22615825 3.880845e-02 0.01209977 0.0000000  0
SI 0.00000000 0.000000e+00 0.02299583 0.3206432  0

# 5. Bootstrap inference ----
res_boot <- csem(dat, model,
                 .approach_weights = "PLS-PM",
                 .disattenuate = TRUE,
                 .resample_method = "bootstrap",
                 .R = 10000, .seed = 42)
infer(res_boot)

$Path_estimates
$Path_estimates$mean
    JS ~ EP     JS ~ AC     OC ~ EP     OC ~ AC     OC ~ JS     SI ~ JS
 0.25498139 -0.00699852  0.43417417  0.18400708  0.10066826  0.13183657
    SI ~ OC
 0.49777868

$Path_estimates$sd
   JS ~ EP    JS ~ AC    OC ~ EP    OC ~ AC    OC ~ JS    SI ~ JS    SI ~ OC
0.05238688 0.06081747 0.05735791 0.04732706 0.05289946 0.04685520 0.05095048

$Path_estimates$bias
      JS ~ EP       JS ~ AC       OC ~ EP       OC ~ AC       OC ~ JS
 0.0101564300 -0.0037164466  0.0053926890  0.0114526664  0.0045988304
      SI ~ JS       SI ~ OC
 0.0006099818  0.0077649037

$Path_estimates$CI_standard_z
       JS ~ EP    JS ~ AC   OC ~ EP    OC ~ AC     OC ~ JS    SI ~ JS   SI ~ OC
95%L 0.1319921 -0.1187657 0.3109694 0.06834241 -0.01221044 0.03878211 0.3823878
95%U 0.3373449  0.1196344 0.5358082 0.25386108  0.19515164 0.22245111 0.5821100

$Path_estimates$CI_standard_t
       JS ~ EP    JS ~ AC   OC ~ EP    OC ~ AC     OC ~ JS    SI ~ JS   SI ~ OC
95%L 0.1316782 -0.1191302 0.3106256 0.06805876 -0.01252749 0.03850128 0.3820824
95%U 0.3376589  0.1199989 0.5361520 0.25414473  0.19546869 0.22273193 0.5824154

$Path_estimates$CI_percentile
       JS ~ EP    JS ~ AC   OC ~ EP    OC ~ AC      OC ~ JS    SI ~ JS
95%L 0.1525445 -0.1247503 0.3226357 0.09125872 -0.004465693 0.03809319
95%U 0.3584825  0.1147835 0.5483708 0.27329485  0.200425662 0.22330878
       SI ~ OC
95%L 0.3987664
95%U 0.5977819
```

```
$Path_estimates$CI_basic
       JS ~ EP    JS ~ AC   OC ~ EP    OC ~ AC     OC ~ JS    SI ~ JS   SI ~ OC
95%L 0.1108546 -0.1139147 0.2984068 0.04890864 -0.01748446 0.03792443 0.3667158
95%U 0.3167926  0.1256190 0.5241419 0.23094477  0.18740689 0.22314002 0.5657313

$Path_estimates$CI_bc
       JS ~ EP    JS ~ AC   OC ~ EP    OC ~ AC     OC ~ JS    SI ~ JS   SI ~ OC
95%L 0.1272011 -0.1176517 0.3130103 0.06447926 -0.01591844 0.03577541 0.3873681
95%U 0.3350133  0.1223213 0.5402480 0.25198518  0.18937667 0.21980757 0.5859620


$Loading_estimates
$Loading_estimates$mean
EP =~ EP1 EP =~ EP2 EP =~ EP3 EP =~ EP4 AC =~ AC1 AC =~ AC2 AC =~ AC3 AC =~ AC4
0.6460326 0.8597909 0.7790753 0.8235464 0.7815283 0.6997112 0.8350940 0.9362963
JS =~ JS1 JS =~ JS2 JS =~ JS3 JS =~ JS4 JS =~ JS5 OC =~ OC1 OC =~ OC2 OC =~ OC3
0.6295581 0.6977020 0.7074801 0.6291963 0.8543881 0.4277283 0.9444227 0.6893385
OC =~ OC4 SI =~ SI1 SI =~ SI2 SI =~ SI3 SI =~ SI4
0.8528464 0.8203070 0.8668863 0.6871950 0.8884818

$Loading_estimates$sd
 EP =~ EP1  EP =~ EP2  EP =~ EP3  EP =~ EP4  AC =~ AC1  AC =~ AC2  AC =~ AC3
0.08569527 0.07032474 0.09069504 0.06417264 0.08339164 0.10445470 0.07696909
 AC =~ AC4  JS =~ JS1  JS =~ JS2  JS =~ JS3  JS =~ JS4  JS =~ JS5  OC =~ OC1
0.04623166 0.11420594 0.10776534 0.10594783 0.12378159 0.08940271 0.07145235
 OC =~ OC2  OC =~ OC3  OC =~ OC4  SI =~ SI1  SI =~ SI2  SI =~ SI3  SI =~ SI4
0.03204257 0.05419105 0.04792753 0.04748498 0.04019511 0.05920661 0.04581260

$Loading_estimates$bias
    EP =~ EP1     EP =~ EP2     EP =~ EP3     EP =~ EP4     AC =~ AC1
 0.0119315350 -0.0188909609  0.0112732015  0.0004773045  0.0123984136
    AC =~ AC2     AC =~ AC3     AC =~ AC4     JS =~ JS1     JS =~ JS2
 0.0231852509  0.0135079989 -0.0570894809  0.0047243690  0.0012432096
    JS =~ JS3     JS =~ JS4     JS =~ JS5     OC =~ OC1     OC =~ OC2
-0.0107473460 -0.0027082401 -0.0460121650  0.0025178269 -0.0130455845
    OC =~ OC3     OC =~ OC4     SI =~ SI1     SI =~ SI2     SI =~ SI3
 0.0034619716 -0.0037206081  0.0032970873 -0.0036331024  0.0093601379
    SI =~ SI4
-0.0074523575

$Loading_estimates$CI_standard_z
     EP =~ EP1 EP =~ EP2 EP =~ EP3 EP =~ EP4 AC =~ AC1 AC =~ AC2 AC =~ AC3
95%L 0.4542099 0.7597389 0.5787699 0.6968157 0.5932869 0.4486132 0.6572213
95%U 0.7901292 1.0354068 0.9342879 0.9483678 0.9201761 0.8580681 0.9589346
     AC =~ AC4 JS =~ JS1 JS =~ JS2 JS =~ JS3 JS =~ JS4 JS =~ JS5 OC =~ OC1
95%L 0.9598629 0.3962698 0.4839994 0.5213208 0.3920053 0.7711864 0.2826487
95%U 1.1410876 0.8439489 0.9064317 0.9366287 0.8772202 1.1216385 0.5627367
     OC =~ OC2 OC =~ OC3 OC =~ OC4 SI =~ SI1 SI =~ SI2 SI =~ SI3 SI =~ SI4
95%L 0.9077116  0.576202 0.7663514 0.7206440 0.7953715 0.5524319 0.8135955
95%U 1.0333161  0.788627 0.9542239 0.9067817 0.9529335 0.7845176 0.9931776

$Loading_estimates$CI_standard_t
```

```
     EP =~ EP1 EP =~ EP2 EP =~ EP3 EP =~ EP4 AC =~ AC1 AC =~ AC2 AC =~ AC3
95%L 0.4536963 0.7593174 0.5782263 0.6964311 0.5927871 0.4479872 0.6567600
95%U 0.7906428 1.0358283 0.9348315 0.9487524 0.9206759 0.8586942 0.9593959
     AC =~ AC4 JS =~ JS1 JS =~ JS2 JS =~ JS3 JS =~ JS4 JS =~ JS5 OC =~ OC1
95%L 0.9595858 0.3955853 0.4833535 0.5206859 0.3912634 0.7706505 0.2822204
95%U 1.1413647 0.8446334 0.9070776 0.9372637 0.8779621 1.1221744 0.5631650
     OC =~ OC2 OC =~ OC3 OC =~ OC4 SI =~ SI1 SI =~ SI2 SI =~ SI3 SI =~ SI4
95%L 0.9075195 0.5758772 0.7660642 0.7203594 0.7951306 0.5520771 0.8133209
95%U 1.0335082 0.7889518 0.9545111 0.9070663 0.9531744 0.7848724 0.9934521

$Loading_estimates$CI_percentile
     EP =~ EP1 EP =~ EP2 EP =~ EP3 EP =~ EP4 AC =~ AC1 AC =~ AC2 AC =~ AC3
95%L 0.4695362 0.7141417 0.5861862 0.6910018 0.6073221 0.4551459 0.6737403
95%U 0.8058722 0.9842858 0.9390766 0.9448089 0.9418083 0.8749444 0.9751209
     AC =~ AC4 JS =~ JS1 JS =~ JS2 JS =~ JS3 JS =~ JS4 JS =~ JS5 OC =~ OC1
95%L 0.8258577 0.3905978 0.4727571 0.4871845 0.3683305 0.6481645 0.2770523
95%U 0.9970881 0.8437585 0.8930565 0.9021730 0.8566478 0.9895487 0.5581800
     OC =~ OC2 OC =~ OC3 OC =~ OC4 SI =~ SI1 SI =~ SI2 SI =~ SI3 SI =~ SI4
95%L 0.8749187 0.5830295 0.7534215 0.7238992 0.7849956 0.5645130 0.7950191
95%U 0.9959027 0.7959996 0.9427467 0.9082782 0.9439578 0.7985734 0.9756005

$Loading_estimates$CI_basic
     EP =~ EP1 EP =~ EP2 EP =~ EP3 EP =~ EP4 AC =~ AC1 AC =~ AC2 AC =~ AC3
95%L 0.4384668 0.8108598 0.5739812 0.7003746 0.5716547 0.4317369 0.6410351
95%U 0.7748028 1.0810039 0.9268716 0.9541817 0.9061410 0.8515354 0.9424157
     AC =~ AC4 JS =~ JS1 JS =~ JS2 JS =~ JS3 JS =~ JS4 JS =~ JS5 OC =~ OC1
95%L  1.103862 0.3964602 0.4973746 0.5557765 0.4125777 0.9032762 0.2872053
95%U  1.275093 0.8496209 0.9176740 0.9707650 0.9008949 1.2446604 0.5683331
     OC =~ OC2 OC =~ OC3 OC =~ OC4 SI =~ SI1 SI =~ SI2 SI =~ SI3 SI =~ SI4
95%L  0.945125 0.5688295 0.7778286 0.7191474 0.8043472 0.5383761 0.8311725
95%U  1.066109 0.7817996 0.9671537 0.9035264 0.9633095 0.7724365 1.0117539

$Loading_estimates$CI_bc
     EP =~ EP1 EP =~ EP2 EP =~ EP3 EP =~ EP4 AC =~ AC1 AC =~ AC2 AC =~ AC3
95%L 0.4395424 0.7477945 0.5522135 0.6827418 0.5665922 0.3459511 0.6321116
95%U 0.7778976 0.9956127 0.9130303 0.9391844 0.9171489 0.8240510 0.9511571
     AC =~ AC4 JS =~ JS1 JS =~ JS2 JS =~ JS3 JS =~ JS4 JS =~ JS5 OC =~ OC1
95%L 0.9876729 0.3654902 0.4470995 0.5060276 0.3590332 0.7517272 0.2593457
95%U 0.9999848 0.8168630 0.8785544 0.9144032 0.8511615 0.9985365 0.5461919
     OC =~ OC2 OC =~ OC3 OC =~ OC4 SI =~ SI1 SI =~ SI2 SI =~ SI3 SI =~ SI4
95%L 0.8975169 0.5770774 0.7551351 0.7154785 0.7916389 0.5362403 0.8117293
95%U 0.9989863 0.7878350 0.9457524 0.8991749 0.9498524 0.7752105 0.9848405


$Weight_estimates
$Weight_estimates$mean
EP <~ EP1 EP <~ EP2 EP <~ EP3 EP <~ EP4 AC <~ AC1 AC <~ AC2 AC <~ AC3 AC <~ AC4
0.2455644 0.3280188 0.2958856 0.3137174 0.2759010 0.2464848 0.2947993 0.3305977
JS <~ JS1 JS <~ JS2 JS <~ JS3 JS <~ JS4 JS <~ JS5 OC <~ OC1 OC <~ OC2 OC <~ OC3
0.2270613 0.2520258 0.2559356 0.2270724 0.3091283 0.1754396 0.3888774 0.2836912
OC <~ OC4 SI <~ SI1 SI <~ SI2 SI <~ SI3 SI <~ SI4
0.3508269 0.2889617 0.3054825 0.2419354 0.3130994
```

```
$Weight_estimates$sd
 EP <~ EP1  EP <~ EP2  EP <~ EP3  EP <~ EP4  AC <~ AC1  AC <~ AC2  AC <~ AC3
0.02942299 0.03357839 0.02745038 0.02594824 0.03037583 0.03440725 0.02806319
 AC <~ AC4  JS <~ JS1  JS <~ JS2  JS <~ JS3  JS <~ JS4  JS <~ JS5  OC <~ OC1
0.01908799 0.03915388 0.03862629 0.04024470 0.04378556 0.03528609 0.02599380
 OC <~ OC2  OC <~ OC3  OC <~ OC4  SI <~ SI1  SI <~ SI2  SI <~ SI3  SI <~ SI4
0.01863876 0.02250749 0.01702047 0.01577287 0.01526872 0.01871205 0.01732339

$Weight_estimates$bias
   EP <~ EP1     EP <~ EP2     EP <~ EP3     EP <~ EP4     AC <~ AC1
 0.0030387273 -0.0080520423  0.0022231350 -0.0010831046  0.0051896131
   AC <~ AC2     AC <~ AC3     AC <~ AC4     JS <~ JS1     JS <~ JS2
 0.0083673086  0.0056248639 -0.0190452433  0.0047250033  0.0042029478
   JS <~ JS3     JS <~ JS4     JS <~ JS5     OC <~ OC1     OC <~ OC2
 0.0003667173  0.0022200357 -0.0112636922  0.0013641970 -0.0030970177
   OC <~ OC3     OC <~ OC4     SI <~ SI1     SI <~ SI2     SI <~ SI3
 0.0029027401  0.0001601336  0.0007293655 -0.0016274889  0.0028024441
   SI <~ SI4
-0.0029766061

$Weight_estimates$CI_standard_z
     EP <~ EP1 EP <~ EP2 EP <~ EP3 EP <~ EP4 AC <~ AC1 AC <~ AC2 AC <~ AC3
95%L 0.1818189 0.2783105 0.2376376 0.2650260 0.2059863 0.1623132 0.2285467
95%U 0.2971549 0.4099353 0.3452411 0.3667413 0.3250573 0.2971872 0.3385524
     AC <~ AC4 JS <~ JS1 JS <~ JS2 JS <~ JS3 JS <~ JS4 JS <~ JS5 OC <~ OC1
95%L 0.3312764 0.1408711 0.1679138 0.1763240 0.1368142 0.2624962 0.1217643
95%U 0.4061000 0.2943515 0.3193261 0.3340803 0.3084504 0.4008151 0.2236581
     OC <~ OC2 OC <~ OC3 OC <~ OC4 SI <~ SI1 SI <~ SI2 SI <~ SI3 SI <~ SI4
95%L 0.3585401 0.2337718 0.3171472 0.2565887 0.2788113 0.1996555 0.2850994
95%U 0.4316027 0.3219996 0.3838662 0.3184173 0.3386636 0.2730054 0.3530058

$Weight_estimates$CI_standard_t
     EP <~ EP1 EP <~ EP2 EP <~ EP3 EP <~ EP4 AC <~ AC1 AC <~ AC2 AC <~ AC3
95%L 0.1816426 0.2781092 0.2374731 0.2648705 0.2058042 0.1621070 0.2283785
95%U 0.2973313 0.4101366 0.3454056 0.3668968 0.3252394 0.2973934 0.3387206
     AC <~ AC4 JS <~ JS1 JS <~ JS2 JS <~ JS3 JS <~ JS4 JS <~ JS5 OC <~ OC1
95%L 0.3311620 0.1406365 0.1676823 0.1760828 0.1365518 0.2622847 0.1216085
95%U 0.4062144 0.2945862 0.3195576 0.3343216 0.3087129 0.4010266 0.2238139
     OC <~ OC2 OC <~ OC3 OC <~ OC4 SI <~ SI1 SI <~ SI2 SI <~ SI3 SI <~ SI4
95%L 0.3584284 0.2336369 0.3170452 0.2564942 0.2787198 0.1995434 0.2849956
95%U 0.4317144 0.3221345 0.3839682 0.3185118 0.3387551 0.2731176 0.3531097

$Weight_estimates$CI_percentile
     EP <~ EP1 EP <~ EP2 EP <~ EP3 EP <~ EP4 AC <~ AC1 AC <~ AC2 AC <~ AC3
95%L 0.1862724 0.2664972 0.2383721  0.266352 0.2161097 0.1666236 0.2393982
95%U 0.3022760 0.3966996 0.3471264  0.369077 0.3392785 0.3061925 0.3497473
     AC <~ AC4 JS <~ JS1 JS <~ JS2 JS <~ JS3 JS <~ JS4 JS <~ JS5 OC <~ OC1
95%L 0.2900372 0.1467359 0.1749539 0.1782516 0.1372702 0.2339649 0.1208154
95%U 0.3632796 0.3002947 0.3264341 0.3359858 0.3123902 0.3696986 0.2220314
     OC <~ OC2 OC <~ OC3 OC <~ OC4 SI <~ SI1 SI <~ SI2 SI <~ SI3 SI <~ SI4
95%L 0.3553286 0.2406968 0.3184753 0.2587451 0.2762347 0.2039864 0.2806058
95%U 0.4280407 0.3305418 0.3845823 0.3195894 0.3366333 0.2783462 0.3488488
```

```
$Weight_estimates$CI_basic
     EP <~ EP1 EP <~ EP2 EP <~ EP3 EP <~ EP4 AC <~ AC1 AC <~ AC2 AC <~ AC3
95%L 0.1766979 0.2915462 0.2357523 0.2626903 0.1917651 0.1533079 0.2173518
95%U 0.2927014 0.4217486 0.3445066 0.3654153 0.3149339 0.2928768 0.3277009
     AC <~ AC4 JS <~ JS1 JS <~ JS2 JS <~ JS3 JS <~ JS4 JS <~ JS5 OC <~ OC1
95%L 0.3740968 0.1349279 0.1608058 0.1744185 0.1328744 0.2936128 0.1233909
95%U 0.4473393 0.2884867 0.3122860 0.3321528 0.3079944 0.4293464 0.2246069
     OC <~ OC2 OC <~ OC3 OC <~ OC4 SI <~ SI1 SI <~ SI2 SI <~ SI3 SI <~ SI4
95%L 0.3621021 0.2252296 0.3164310 0.2554166 0.2808416 0.1943147 0.2892564
95%U 0.4348143 0.3150746 0.3825381 0.3162609 0.3412402 0.2686745 0.3574994

$Weight_estimates$CI_bc
     EP <~ EP1 EP <~ EP2 EP <~ EP3 EP <~ EP4 AC <~ AC1 AC <~ AC2 AC <~ AC3
95%L 0.1781986 0.2817962 0.2311247 0.2708809 0.2074401 0.1309097 0.2290425
95%U 0.2950485 0.4179810 0.3407272 0.3750284 0.3286914 0.2865969 0.3393959
     AC <~ AC4 JS <~ JS1 JS <~ JS2 JS <~ JS3 JS <~ JS4 JS <~ JS5 OC <~ OC1
95%L 0.3349048 0.1322677 0.1609181 0.1775223 0.1301494 0.2548716 0.1107709
95%U 0.4126115 0.2896260 0.3139789 0.3346152 0.3051568 0.3866948 0.2161321
     OC <~ OC2 OC <~ OC3 OC <~ OC4 SI <~ SI1 SI <~ SI2 SI <~ SI3 SI <~ SI4
95%L 0.3623654 0.2365131 0.3180438 0.2570640 0.2808688 0.1971703 0.2867658
95%U 0.4397117 0.3242966 0.3841103 0.3181764 0.3420768 0.2708492 0.3562541


$Exo_construct_correlation
$Exo_construct_correlation$mean
 EP ~~ AC
0.2608173

$Exo_construct_correlation$sd
  EP ~~ AC
0.05493887

$Exo_construct_correlation$bias
   EP ~~ AC
0.005870009

$Exo_construct_correlation$CI_standard_z
      EP ~~ AC
95%L 0.1413991
95%U 0.3567555

$Exo_construct_correlation$CI_standard_t
      EP ~~ AC
95%L 0.1410698
95%U 0.3570848

$Exo_construct_correlation$CI_percentile
      EP ~~ AC
95%L 0.1489720
95%U 0.3655552

$Exo_construct_correlation$CI_basic
      EP ~~ AC
```

```
95%L 0.1325994
95%U 0.3491826

$Exo_construct_correlation$CI_bc
      EP ~~ AC
95%L 0.1269960
95%U 0.3494058


$Indirect_effect
$Indirect_effect$mean
      OC ~ EP       OC ~ AC       SI ~ EP       SI ~ AC       SI ~ JS
 0.0255739207 -0.0008060915  0.2637702742  0.0906133368  0.0501359641

$Indirect_effect$sd
   OC ~ EP    OC ~ AC    SI ~ EP    SI ~ AC    SI ~ JS
0.01464718 0.00685761 0.04074460 0.02905960 0.02697351

$Indirect_effect$bias
      OC ~ EP       OC ~ AC       SI ~ EP       SI ~ AC       SI ~ JS
 0.0020537265 -0.0004907846  0.0100086766  0.0066444973  0.0030606203

$Indirect_effect$CI_standard_z
          OC ~ EP     OC ~ AC   SI ~ EP    SI ~ AC      SI ~ JS
95%L -0.007241482 -0.01326519 0.1638950 0.02036857 -0.008852386
95%U  0.050174418  0.01361615 0.3236109 0.13428011  0.096881833

$Indirect_effect$CI_standard_t
          OC ~ EP     OC ~ AC   SI ~ EP    SI ~ AC     SI ~ JS
95%L -0.007329269 -0.01330629 0.1636508 0.02019441 -0.00901405
95%U  0.050262205  0.01365725 0.3238551 0.13445428  0.09704350

$Indirect_effect$CI_percentile
          OC ~ EP     OC ~ AC   SI ~ EP    SI ~ AC      SI ~ JS
95%L -0.001161769 -0.01616362 0.1901888 0.03657312 -0.002077644
95%U  0.056265783  0.01274064 0.3471311 0.14899316  0.102420267

$Indirect_effect$CI_basic
         OC ~ EP     OC ~ AC   SI ~ EP     SI ~ AC     SI ~ JS
95%L -0.01333285 -0.01238969 0.1403747 0.005655529 -0.01439082
95%U  0.04409470  0.01651457 0.2973171 0.118075566  0.09010709

$Indirect_effect$CI_bc
          OC ~ EP     OC ~ AC   SI ~ EP    SI ~ AC      SI ~ JS
95%L -0.003885354 -0.01580347 0.1758929 0.02554995 -0.008325499
95%U  0.052323822  0.01311098 0.3308011 0.13660292  0.096759834


$Total_effect
$Total_effect$mean
    JS ~ EP     JS ~ AC     OC ~ EP     OC ~ AC     OC ~ JS     SI ~ EP
 0.25498139 -0.00699852  0.45974809  0.18320099  0.10066826  0.26377027
    SI ~ AC     SI ~ JS     SI ~ OC
```

```
 0.09061334  0.18197253  0.49777868

$Total_effect$sd
   JS ~ EP    JS ~ AC    OC ~ EP    OC ~ AC    OC ~ JS    SI ~ EP    SI ~ AC 
0.05238688 0.06081747 0.05499662 0.04785428 0.05289946 0.04074460 0.02905960 
   SI ~ JS    SI ~ OC 
0.04375639 0.05095048 

$Total_effect$bias
     JS ~ EP      JS ~ AC      OC ~ EP      OC ~ AC      OC ~ JS      SI ~ EP 
 0.010156430 -0.003716447  0.007446416  0.010961882  0.004598830  0.010008677 
     SI ~ AC      SI ~ JS      SI ~ OC 
 0.006644497  0.003670602  0.007764904 

$Total_effect$CI_standard_z
       JS ~ EP    JS ~ AC   OC ~ EP    OC ~ AC     OC ~ JS   SI ~ EP    SI ~ AC
95%L 0.1319921 -0.1187657 0.3370639 0.06748455 -0.01221044 0.1638950 0.02036857
95%U 0.3373449  0.1196344 0.5526467 0.25506989  0.19515164 0.3236109 0.13428011
        SI ~ JS   SI ~ OC
95%L 0.08887038 0.3823878
95%U 0.26039228 0.5821100

$Total_effect$CI_standard_t
       JS ~ EP    JS ~ AC   OC ~ EP    OC ~ AC     OC ~ JS   SI ~ EP    SI ~ AC
95%L 0.1316782 -0.1191302 0.3367343 0.06719774 -0.01252749 0.1636508 0.02019441
95%U 0.3376589  0.1199989 0.5529763 0.25535671  0.19546869 0.3238551 0.13445428
        SI ~ JS   SI ~ OC
95%L 0.08860813 0.3820824
95%U 0.26065453 0.5824154

$Total_effect$CI_percentile
       JS ~ EP    JS ~ AC   OC ~ EP  OC ~ AC      OC ~ JS   SI ~ EP    SI ~ AC
95%L 0.1525445 -0.1247503 0.3533574 0.090650 -0.004465693 0.1901888 0.03657312
95%U 0.3584825  0.1147835 0.5683438 0.272157  0.200425662 0.3471311 0.14899316
        SI ~ JS   SI ~ OC
95%L 0.09737975 0.3987664
95%U 0.26719106 0.5977819

$Total_effect$CI_basic
       JS ~ EP    JS ~ AC   OC ~ EP    OC ~ AC     OC ~ JS   SI ~ EP
95%L 0.1108546 -0.1139147 0.3213667 0.05039744 -0.01748446 0.1403747
95%U 0.3167926  0.1256190 0.5363531 0.23190445  0.18740689 0.2973171
         SI ~ AC   SI ~ JS   SI ~ OC
95%L 0.005655529 0.0820716 0.3667158
95%U 0.118075566 0.2518829 0.5657313

$Total_effect$CI_bc
       JS ~ EP    JS ~ AC   OC ~ EP    OC ~ AC     OC ~ JS   SI ~ EP    SI ~ AC
95%L 0.1272011 -0.1176517 0.3393091 0.06572548 -0.01591844 0.1758929 0.02554995
95%U 0.3350133  0.1223213 0.5556759 0.25411065  0.18937667 0.3308011 0.13660292
        SI ~ JS   SI ~ OC
95%L 0.09090796 0.3873681
95%U 0.26100956 0.5859620
```

```
attr(,"class")
[1] "cSEMInfer"

# 6. Model fit ----
testOMF(res, .R = 10000, .seed = 42,
        .handle_inadmissibles = "replace")

--------------------------------------------------------------------------------
--------- Test for overall model fit based on Beran & Srivastava (1985) --------

Null hypothesis:

       +-------------------------------------------------------------------+
       |                                                                   |
       |   H0: The model-implied indicator covariance matrix equals the   |
       |   population indicator covariance matrix.                         |
       |                                                                   |
       +-------------------------------------------------------------------+

Test statistic and critical value:

                                    Critical value
 Distance measure    Test statistic   95%
 dG                      0.3253     0.3554
 SRMR                    0.0706     0.0505
 dL                      1.1505     0.5898
 dML                     1.8972     2.1208


Decision:

                      Significance level
 Distance measure          95%
 dG                    Do not reject
 SRMR                      reject
 dL                        reject
 dML                   Do not reject

Additional information:

 Out of 19853 bootstrap replications 10000 are admissible.
 See ?verify() for what constitutes an inadmissible result.

 The seed used was: 42
--------------------------------------------------------------------------------
```

References

Barba, L. A. (2022). Defining the role of open source software in research reproducibility. *Computer*, *55*(8), 40–48. doi: 10.1109/mc.2022.3177133

Benitez, J., Henseler, J., Castillo, A., & Schuberth, F. (2020). How to perform and report an impactful analysis using partial least squares: Guidelines for confirmatory and explanatory IS research. *Information & Management*, *2*(57), 103168. doi: 10.1016/j.im.2019.05.003

Browne, M. W. (1974). Generalized least squares estimators in the analysis of covariance structures. *South African Statistical Journal*, *8*(1), 1–24. doi: 10.1002/j.2333-8504.1973.tb00197.x

Cheah, J.-H., Magno, F., & Cassia, F. (2024). Reviewing the SmartPLS 4 software: the latest features and enhancements. *Journal of Marketing Analytics*, *12*, 97–102. doi: 10.1057/s41270-023-00266-y

Cheah, J.-H., Sarstedt, M., Hair, J. F., & Ringle, C. M. (2026). Consistent partial least squares structural equation modeling using smartpls. *Structural Equation Modeling: A Multidisciplinary Journal*, 1–14. doi: 10.1080/10705511.2026.2633754

Chuah, F., Memon, M. A., Thurasamy, R., Cheah, J.-H., Ting, H., & Cham, T. H. (2021, aug). PLS-SEM using R: An introduction to cSEM and SEMinR. *Journal of Applied Structural Equation Modeling*, *5*(2), 1–35. doi: 10.47263/jasem.5(2)01

Cook, R. D., & Forzani, L. (2023). On the role of partial least squares in path analysis for the social sciences. *Journal of Business Research*, *167*, 114132. doi: 10.1016/j.jbusres.2023.114132

Devlieger, I., & Rosseel, Y. (2017). Factor score path analysis. *Methodology*, *13*, 31-38.

Dijkstra, T. K. (1985). *Latent variables in linear stochastic models: Reflections on "maximum likelihood" and "partial least squares" methods* (Vol. 1). Amsterdam: Sociometric Research Foundation.

Dijkstra, T. K. (2013). *A note on how to make pls consistent* (Tech. Rep.). working paper, University of Groningen, Groningen, The Netherlands. Retrieved from `http://www.rug.nl/staff/tkdijkstra/how-to-make-pls-consistent.pdf`

Dijkstra, T. K., & Henseler, J. (2015a). Consistent and asymptotically normal PLS estimators for linear structural equations. *Computational Statistics & Data Analysis*, *81*, 10–23. doi: 10.1016/j.csda.2014.07.008

Dijkstra, T. K., & Henseler, J. (2015b). Consistent partial least squares path modeling. *MIS Quarterly*, *39*(2), 297–316. doi: 10.25300/MISQ/2015/39.2.02

Dijkstra, T. K., & Schermelleh-Engel, K. (2014). Consistent partial least squares for nonlinear structural equation models. *Psychometrika*, *79*(4), 585–604. doi: 10.1007/s11336-013-9370-0

Evermann, J., & Rönkkö, M. (2023). Recent developments in PLS. *Communications of the Association for Information Systems*, *52*, 663–667. doi: 10.17705/1CAIS.05229

Franke, G., & Sarstedt, M. (2019). Heuristics versus statistics in discriminant validity testing: a comparison of four procedures. *Internet Research*, *29*(3), 430–447. doi: 10.1108/IntR-12-2017-0515

Hair, J. F., Black, W. C., Babin, B. J., & Anderson, R. E. (2019). *Multivariate data analysis* (8th ed.). Cengage.

Hair, J. F., Hult, G. T. M., Ringle, C. M., & Sarstedt, M. (2026). *A primer on partial least squares structural equation modeling (PLS-SEM)* (4th ed.). Thousand Oaks, CA: Sage.

Hair, J. F., Ringle, C. M., & Sarstedt, M. (2011). PLS-SEM: Indeed a silver bullet. *Journal of Marketing Theory and Practice*, *19*(2), 139–152. doi: 10.2753/MTP1069-6679190202

Henseler, J. (2025). Adanco 2.4 [Computer software manual]. Enschede.

Henseler, J., Ringle, C. M., & Sarstedt, M. (2015). A new criterion for assessing discriminant validity in variance-based structural equation modeling. *Journal of the Academy of Marketing Science*, *43*(1), 115–135. doi: 10.1007/s11747-014-0403-8

Hu, L.-t., & Bentler, P. M. (1999). Cutoff criteria for fit indexes in covariance structure analysis: Conventional criteria versus new alternatives. *Structural Equation Modeling: A Multidisciplinary Journal*, *6*(1), 1–55. doi: 10.1080/10705519909540118

Hui, B. S., & Wold, H. (1982). Consistency and consistency at large of partial least squares estimates. In K. G. Jöreskog & H. Wold (Eds.), *Systems under indirect observation: Causality, structure, prediction part ii* (pp. 119–130). Amsterdam: North-Holland.

Humble, J. (2024). plspm: A library implementing partial least squares path modeling [Computer software manual]. Retrieved from `https://plspm.readthedocs.io/` (Python package version 0.5.7)

Hwang, H., Cho, G., Jung, K., Falk, C., Flake, J., Jin, M. J., & Lee, S. H. (2021). An approach to structural equation modeling with both factors and components: Integrated generalized structured component analysis. *Psychological Methods*, *26*(3), 273–294. doi: 10.1037/met0000336

Hwang, H., & Takane, Y. (2004). Generalized structured component analysis. *Psychometrika*, *69*(1), 81–99. doi: 10.1007/BF02295841

Ince, D. C., Hatton, L., & Graham-Cumming, J. (2012). The case for open computer programs. *Nature*, *482*(7386), 485–488. doi: 10.1038/nature10836

Jöreskog, K. G. (1970). A general method for estimating a linear structural equation system. *ETS Research Bulletin Series*, *1970*(2), i–41. doi: 10.1002/j.2333-8504.1970.tb00783.x

Jung, S., & Park, J. (2018). Consistent partial least squares path modeling via regularization. *Frontiers in Psychology*, *9*. doi: 10.3389/fpsyg.2018.00174

Kline, R. B. (2023). *Principles and practice of structural equation modeling* (5th ed.). New York: The Guilford Press.

Kock, N. (2025). *WarpPLS user manual: Version 8.0*. Laredo, TX: ScriptWarp Systems.

Marcoulides, G. A., & Schumacker, R. E. (Eds.). (1996). *Advanced structural equation modeling: Issues and techniques*. New York: Psychology Press.

Morin, A., Urban, J., Adams, P. D., Foster, I., Sali, A., Baker, D., & Sliz, P. (2012). Shining light into black boxes. *Science*, *336*(6078), 159–160. doi: 10.1126/science.1218263

Müller, T., Schuberth, F., & Henseler, J. (2018). PLS path modeling - a confirmatory approach to study tourism technology and tourist behavior. *Journal of Hospitality and Tourism Technology*, *9*(3), 249–266. doi: 10.1108/JHTT-09-2017-0106

Python Software Foundation. (2024). Python language reference, version 3.12 [Computer software manual]. Retrieved from `https://www.python.org`

R Core Team. (2025). R: A language and environment for statistical computing [Computer software manual]. Vienna, Austria. Retrieved from `https://www.R-project.org/`

Rademaker, M. E., & Schuberth, F. (2020). csem: Composite-based structural equation modeling [Computer software manual]. Retrieved from `https://floschuberth.github.io/cSEM/` (Package version: 0.6.1.9000)

Ray, S., Danks, N. P., & Calero Valdez, A. (2026). seminr: Building and estimating structural equation models [Computer software manual]. Retrieved from `https://CRAN.R-project.org/package=seminr` (R package version 2.4.2) doi: 10.32614/CRAN.package.seminr

Ringle, C. M., Sarstedt, M., Mitchell, R., & Gudergan, S. P. (2020, jan). Partial least squares structural equation modeling in HRM research. *The International Journal of Human Resource Management*, *31*(12), 1617–1643. doi: 10.1080/09585192.2017.1416655

Ringle, C. M., Wende, S., & Becker, J.-M. (2024). *SmartPLS 4*. Boenningstedt: SmartPLS GmbH. Retrieved from `https://www.smartpls.com`

Rönkkö, M. (2022). matrixpls: Matrix-based partial least squares estimation [Computer software manual]. Retrieved from `https://github.com/mronkko/matrixpls` (R package

version 1.0.15)
Rönkkö, M., & Evermann, J. (2013). A critical examination of common beliefs about partial least squares path modeling. *Organizational Research Methods*, *16*(13), 425–448. doi: 10.1177/1094428112474693
Rosseel, Y. (2012). lavaan: An R package for structural equation modeling. *Journal of Statistical Software*, *48*(2), 1–36. doi: 10.18637/jss.v048.i02
Rosseel, Y., Jorgensen, T. D., & De Wilde, L. (2025). lavaan: Latent variable analysis [Computer software manual]. Retrieved from `https://CRAN.R-project.org/package=lavaan` (R package version 0.6-20) doi: 10.32614/CRAN.package.lavaan
Sanchez, G., Trinchera, L., Russolillo, G., & Bertrand, F. (2025). Partial least squares path modeli (PLS-PM, Frederic) [Computer software manual]. Retrieved from `https://CRAN.R-project.org/package=plspm` (R package version 0.6.0) doi: 10.32614/CRAN.package.plspm
Sarstedt, M., Adler, S. J., Ringle, C. M., Cho, G., Diamantopoulos, A., Hwang, H., & Liengaard, B. D. (2024). Same model, same data, but different outcomes: Evaluating the impact of method choices in structural equation modeling. *Journal of Product Innovation Management*, *41*(6), 1100–1117. doi: 10.1111/jpim.12738
Sarstedt, M., Hair, J. F., Pick, M., Liengaard, B. D., Radomir, L., & Ringle, C. M. (2022). Progress in partial least squares structural equation modeling use in marketing research in the last decade. *Psychology & Marketing*, *39*, 1035–1064. doi: 10.1002/mar.21640
Schamberger, T., Schuberth, F., Henseler, J., & Dijkstra, T. K. (2020). Robust partial least squares path modeling. *Behaviormetrika*, *47*, 307–334. doi: 10.1007/s41237-019-00088-2
Schuberth, F., Henseler, J., & Dijkstra, T. K. (2018). Partial least squares path modeling using ordinal categorical indicators. *Quality & Quantity*, *52*(1), 9–35. doi: 10.1007/s11135-016-0401-7
Schuberth, F., Rosseel, Y., Rönkkö, M., Trinchera, L., Kline, R. B., & Henseler, J. (2023). Structural parameters under partial least squares and covariance-based structural equation modeling: A comment on Yuan and Deng (2021). *Structural Equation Modeling: A Multidisciplinary Journal*, *30*(3), 339–345. doi: 10.1080/10705511.2022.2134140
Schuberth, F., Zaza, S., & Henseler, J. (2023). Partial least squares is an estimator for structural equation models: A comment on Evermann and Rönkkö (2021). *Communications of the Association for Information Systems*, *52*, 711–729. doi: 10.17705/1CAIS.05232
Shmueli, G., Sarstedt, M., Hair, J. F., Cheah, J.-H., Ting, H., Vaithilingam, S., & Ringle, C. M. (2019). Predictive model assessment in PLS-SEM: guidelines for using PLSpredict. *European Journal of Marketing*, *53*(11), 2322–2347. doi: 10.1108/EJM-02-2019-0189
Skrondal, A., & Laake, P. (2001). Regression among factor scores. *Psychometrika*, *66*(4), 563–575. doi: 10.1007/BF02296196
UNESCO. (2021). *UNESCO recommendation on open science*. Paris: UNESCO. doi: 10.54677/MNMH8546
Venturini, S., Mehmetoglu, M., & Latan, H. (2023). Software packages forÂ partial least squares structural equation modeling: An updated review. In *Partial least squares path modeling* (pp. 113–152). Springer International Publishing. doi: 10.1007/978-3-031-37772-3_5
Wold, H. (1982). Soft modeling: The basic design and some extensions. In K. G. Jöreskog & H. Wold (Eds.), *Systems under indirect observation: Causality, structure, prediction part ii* (pp. 1–54). Amsterdam: North-Holland.